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An attention-guided hybrid statistical and deep learning modeling for enhanced time series forecasting: A case study of South African telecommunication companies

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ABSTRACT

Accurate stock volatility forecasting is critical for informed investment decisions and effective risk management. This study proposes an attention-guided hybrid modeling framework that integrates Generalized Autoregressive Conditional Heteroskedasticity (GARCH) and Exponential GARCH (EGARCH) models with Long Short-Term Memory (LSTM) networks to improve volatility prediction for South African telecommunication companies (MTN SA, Telkom SA, and Vodacom SA). While GARCH-based models capture volatility clustering and asymmetries and LSTMs learn non-linear dependencies, each method on its own has limitations. We therefore develop GARCH-LSTM and EGARCH-LSTM hybrids enhanced with attention mechanisms that dynamically weight time-series features. The models are trained on one year of stock price data (July 2023–July 2024), incorporating technical indicators such as Simple Moving Average (SMA), Exponential Moving Average (EMA), and Relative Strength Index (RSI) to enrich the feature sets. Using an 80/20 train–test split and rolling window validation, performance is evaluated via Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE). The attention-enhanced EGARCH-LSTM achieves the lowest MAE values: 0.0780 (MTN), 0.0901 (Telkom), and 0.0905 (Vodacom). For RMSE, the lowest errors are model-specific: LSTM-Attn performs best for MTN (0.1036), GARCH-LSTM (without attention) for Telkom (0.1190), and EGARCH-LSTM-Attn for Vodacom (0.1259). These models reduce forecasting errors by up to 4.84% in MAE and 3.67% in RMSE compared to non-attention counterparts. Volatility projections show Telkom exhibits the most significant upward volatility trend rising from 2.66 to 2.99 (GARCH) and from 0.0249 to 0.0306 (EGARCH) over 10 days, while MTN and Vodacom remain more stable. Correlation analysis confirms that attention-based models provide more consistent forecasts across stocks. These findings suggest that combining econometric volatility models with deep learning (DL) and attention mechanisms yields a robust forecasting strategy, particularly well-suited for volatile emerging markets.

Introduction

Understanding and forecasting stock market volatility is fundamental to investment strategies, risk management, and maintaining market stability. In emerging economies like South Africa, where major telecommunication companies such as Telkom SA, MTN SA,

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and Vodacom SA play an important role in economic growth [1], the ability to accurately predict volatility is crucial. Stock volatility impacts both individual and institutional investment decisions, particularly in sectors with high market sensitivity. This work focuses on volatility forecasting for these three leading SA telecommunication firms, aiming to provide insights into better decision-making tools for analysts and investors.

Traditional econometric models, such as the Generalized (ARCH) GARCH model and its extensions, including the EGARCH, have been widely used to model financial time-series volatility [2,3]. GARCH models effectively capture volatility clustering, a pattern in which phases of high volatility tend to occur consecutively [4], while EGARCH extends this by modeling asymmetric market reactions, where negative shocks amplify volatility more than equally sized positive shocks [5]. However, these models struggle to capture complex, non-linear dependencies and dynamic market conditions, especially in emerging markets prone to external shocks [6]. To overcome these limitations, machine learning (ML) techniques have gained prominence owing to their capacity to detect non-linear relationships in financial data [7]. Among these, LSTM networks have been particularly effective in financial time-series forecasting owing to their capacity to model long-term dependencies [8]. Studies show that ML models, including random forests (RF), support vector machines (SVMs), and deep neural networks, often outperform traditional econometric models in predictive accuracy [9,10], but they present challenges such as interpretability issues, inefficient weighting of time steps, and computational complexity [11].

To address these challenges, this work incorporates an attention mechanism to enhance volatility forecasting. Attention mechanisms have gained popularity in time-series modeling by dynamically assigning importance weights to past observations, allowing models to focus on key volatility drivers such as financial crises or earnings announcements [12,13]. Instead of treating all historical data equally, the attention mechanism selectively emphasizes time steps that have a significant influence on future predictions. This approach has been successfully applied in stock market prediction [14] and risk modeling, demonstrating its effectiveness in refining DL models by reducing noise, improving accuracy, and enhancing interpretability. By integrating attention, this work aims to improve model efficiency in capturing volatility patterns without introducing unnecessary computational overhead.

The aim of this paper is to enhance volatility forecasting for South African telecommunication stocks by developing and evaluating models that integrate statistical and ML techniques. Specifically, we assess the performance of six different models: traditional statistical models, DL approaches, hybrid models, and attention-enhanced versions by comparing their predictive accuracy. The work contributes to the growing body of research on hybrid financial models, providing insights applicable to emerging markets, where traditional models often fail to capture heightened volatility and external shocks [1,6]. Enhanced volatility projections from this research are expected to aid financial analysts, investors, and policymakers in making more informed decisions, ultimately contributing to improved market stability and risk management [14–16].

The rest of this paper is organized as follows: Section "Background and related work" reviews the background on traditional econometric and ML-based approaches to volatility forecasting. Section "Methodology" details the methodology, including data collection, preprocessing, and model implementation. Section "Evaluation Metrics" outlines the evaluation metrics used to assess model performance. Section "EGARCH estimation" presents the experimental results, followed by Section "Discussion", which interprets the findings, discussing trade-offs between model complexity, accuracy, and computational efficiency. Finally, Section "Conclusion" wraps up the paper by highlighting key findings and suggesting potential avenues for future research.

Background and related work

Volatility forecasting in financial markets has long relied on econometric models, with the GARCH model introduced in [3] being a cornerstone in financial time series analysis. GARCH models effectively capture volatility clustering, a phenomenon where periods of high volatility are followed by similarly volatile periods [2,4]. However, traditional GARCH models assume symmetric effects of positive and negative shocks, which can limit their ability to capture real-world asymmetries, especially in markets where negative shocks often have a greater impact on volatility than positive ones. To address this, extensions like the EGARCH model [5] allow for asymmetric volatility, making them more suitable for markets with frequent fluctuations, such as those in emerging economies [6]. Other variants, including Threshold GARCH (TGARCH), have also been developed to capture regime-dependent volatility dynamics [17].

Despite their effectiveness in capturing statistical properties such as volatility clustering and asymmetric shocks, GARCH models struggle to model the complex, non-linear relationships often present in financial markets [15,18]. This limitation has led to the increased use of ML models, which are capable of capturing non-linear patterns and adapting to high-dimensional data. Techniques such as RF [19], SVM [20], and DL models, including LSTM networks [8], have shown great promise in time series analysis. LSTM networks, a specialized form of recurrent neural network, are known for their effectiveness in forecasting financial time series due to their capability to learn long-term dependencies and adjust to changing market conditions. For example, [10] demonstrated that DL models outperformed traditional econometric models in predicting stock price volatility by identifying complex patterns in the data. However, ML models, particularly DL models, frequently face criticism for their lack of interpretability [11], which can make them less appealing in financial settings where transparency is crucial.

To overcome the limitations of both traditional econometric models and ML techniques, hybrid models that combine the strengths of both approaches have emerged. These models integrate the stability and interpretability of GARCH with the flexibility and predictive power of ML. Recent studies have highlighted the success of such hybrid models in forecasting volatility. For example, [15] combined GARCH with artificial neural networks (ANNs) to predict gold price volatility, showing improved accuracy over standalone models. Similarly, [14] proposed a hybrid model that integrated GARCH with deep neural networks for financial market prediction, achieving superior forecasting performance. In a similar vein, [21] employed a GARCH-LSTM framework for carbon price

Table 1
Extended comparison of related works on financial volatility forecasting using GARCH-family and DL hybrid models.

Year	Ref.	Model type	GARCH variant	DL method	Hybrid	Application domain	Evaluation metrics
2018	[22]	Hybrid	GARCH, EGARCH, GJR	LSTM	Yes	Stock index volatility	RMSE, MAE, Accuracy
2021	[23]	Hybrid	GARCH	LSTM	Yes	Covariance matrix forecasting	RMSE, volatility spread
2022	[24]	Hybrid	GARCH-type family	LSTM	Yes	Commodity market volatility	MAE, RMSE
2023	[25]	Hybrid	GARCH	GRU	Yes	Risk/volatility forecasting for S&P 500, Bitcoin, gold	RMSE, MAPE
2023	[26]	Hybrid	GARCH	Deep Feedforward Neural Network (DFFNN), LSTM	Yes	Cryptocurrency market	MAE, RMSE
2024	[27]	Hybrid	GARCH	LSTM	Yes	S&P 500 with VIX signals	RMSE, MAPE
2024	[28]	Hybrid	GARCH-like parametric integration	Neural Network (custom)	Yes	General volatility prediction	MAE, model interpretability
2024	[29]	Hybrid	GARCH	LSTM, (GINN)	Yes	Financial market volatility	RMSE, directional accuracy
2025	[30]	Hybrid	GARCH	LSTM, (BiLSTM)	Yes	Turkish BIST30 stock market	MAE, RMSE
2025	[31]	Hybrid	GARCH	Deep Q-Network (RL)	Yes	Value-at-Risk (VaR) estimation	VaR accuracy, capital reduction
2025	**Our work**	Hybrid	GARCH, EGARCH	LSTM + Attention	Yes	SA telecommunication Equity Volatility (MTN, Telkom, Vodacom)	MAE, MSE, RMSE

forecasting and found it to be significantly more accurate than other approaches. Hybrid models have been particularly successful in emerging markets, where traditional models often fail to fully capture the volatility and external shocks that influence asset prices.

For instance, the MULTI-GARCH-LSTM [32] mixed model has demonstrated effectiveness in capturing multi-scale volatility dynamics in stock index futures, highlighting the value of hierarchical GARCH structures enhanced by LSTM's temporal modeling capabilities. Similarly, the Transformer-LSTM-GARCH hybrid model [33] applied to carbon futures markets illustrates the benefits of attention-based mechanisms for long-sequence learning in volatile, climate-sensitive financial instruments. Another innovation involves the integration of GARCH with GRU (Gated Recurrent Unit) networks (GARCH-GRU) [34], enabling efficient learning of sequential dependencies and volatility clustering with reduced computational overhead compared to LSTM. This model proves particularly useful in forecasting abrupt shifts in financial markets. From a methodological standpoint, studies transitioning "From GARCH to Neural Networks" [28] emphasize the conceptual bridge between traditional volatility modeling and neural architectures. These studies argue for embedding GARCH constraints within DL models to preserve interpretability while benefiting from flexible function approximation.

Case-specific implementations, such as a study forecasting stock volatility using a hybrid of DL and GARCH family models on the BIST30 index [30], underscore the empirical effectiveness of these hybrid frameworks in emerging markets. Likewise, an integrated approach using GARCH-CNN-LSTM for stock market volatility prediction showcases how spatial feature extraction (via CNN) and sequential learning (via LSTM) can be combined with statistical properties modeled by GARCH to significantly improve volatility forecasting performance. These studies collectively affirm the growing consensus that hybrid models especially those combining GARCH with LSTM, GRU, CNN, and Transformer architectures, are becoming essential tools in modern financial forecasting. They not only capture the memory and asymmetry properties fundamental to market behavior but also adapt to complex, high-frequency financial data where traditional models alone fall short. In recent years, the combination of GARCH-family models with ML techniques has gained traction as a promising approach for volatility forecasting. Table A.1 outlines a conceptual comparison between GARCH, EGARCH, LSTM, and hybrid models, highlighting their respective strengths and limitations. Hybrid models, in particular, offer a compelling balance by capturing both statistical and non-linear temporal dependencies, although they often require greater computational resources. To support this trend, several empirical studies have demonstrated the superiority of hybrid models across diverse financial domains. Table A.2 provides a summary of key studies that integrate GARCH-type models with DL architectures such as LSTM, GRU, or feedforward networks. These studies report notable improvements in predictive accuracy and risk modeling capabilities. For a more detailed technical comparison, Table 1 presents an extended analysis of recent hybrid modeling approaches. These span applications in stock indices, commodities, cryptocurrencies, and Value-at-Risk estimation. While most studies leverage DL methods in conjunction with traditional GARCH variants, relatively few incorporate attention mechanisms

or focus on emerging market equities. Our work contributes to this growing body of literature by proposing a hybrid EGARCH-LSTM model enhanced with an attention mechanism and supported by technical indicators. Unlike previous studies, we apply this model to the daily volatility of South African telecommunication companies, an underexplored yet economically significant market. This approach not only improves predictive performance but also enhances interpretability, making it valuable for both academic research and practical financial forecasting.

Methodology

The approach used to create the hybrid model with attention for predicting stock volatility in the South African telecommunication industry, with a particular emphasis on Telkom SA, MTN SA, and Vodacom SA, is described in this section. To leverage the strengths of both statistical and DL method, the methodology integrates ML techniques with the conventional GARCH model, enhanced by an attention mechanism to improve predictive accuracy. The process consists of data gathering, model specification, hybrid model construction, and model evaluation.

Over all pipeline

This investigation comprises seven distinct stages as illustrated in Fig. 1, each of which is indispensable for the analysis of stock price volatility pertaining to South African telecommunications firms MTN, Telkom, and Vodacom. Step 1: Data Collection entails the compilation of historical stock price records, encompassing essential variables such as date, opening price, highest price, lowest price, closing price, adjusted closing price, and trading volume. This dataset encompasses a one-year period and serves as the foundational basis for volatility analysis. Step 2: Data Preprocessing is dedicated to the transformation of the raw dataset through the computation of daily returns, normalization of values, and execution of exploratory analyses to identify trends, anomalies, and absent data. Step 3: Model Specification, we delineate econometric and ML frameworks such as GARCH, EGARCH, and LSTM, aimed at capturing volatility patterns and non-linear interdependencies, along with hybrid models like GARCH-LSTM and EGARCH-LSTM with attention mechanism that integrate both methodologies for improved efficacy. Step 4: Feature Engineering, novel features are generated from the raw dataset, including residuals and conditional variances derived from GARCH and EGARCH models, as well as lagged returns and technical indicators intended for application in LSTM models. Step 5: Model Training & Validation involves the education of the models utilizing historical datasets, implementing rolling window validation for time-series data, and employing grid-search techniques for the optimization of hyperparameters. In Step 6: Evaluation Metrics, we evaluate model efficacy by computing various performance metrics such as MAE, MSE and RMSE. Based on these evaluative metrics, one arrives at the following in Ultimately, Step 7: Prediction, employs the chosen model to project future volatility for MTN, Telkom, and Vodacom, thereby furnishing invaluable insights for investors and analysts in order to assess potential risks associated with their investment strategies. For volatility modeling, we used GARCH and EGARCH models to capture the time-varying volatility patterns of stock returns. These econometric models were trained on the historical data, with GARCH focusing on volatility clustering, while EGARCH allowed for modeling asymmetries in the volatility. The LSTM the model was trained with the training data, with hyperparameter tuning based on the validation set. After training, the models were assessed based on the test set to assess their forecasting performance.

Data collection

The data utilized in this work consists of intraday stock prices for three major South African telecommunication companies: Telkom, MTN, and Vodacom from yahoo finance [35]. The data covers a one-year period from July 17, 2023, to July 17, 2024 as shown in Fig. 2. This period is particularly significant due to the dynamic market conditions in the telecommunications sector, driven by ongoing technological advancements and shifts in regulatory frameworks. Throughout the duration, MTN's stock price has shown itself to be the most valuable of the three companies. Vodacom's share price is significantly lower than MTN's, with a more consistent annual growth. It appears to be more stable than MTN, with fewer observable fluctuations. In comparison to Telkom, the two firms with the greatest share prices are Vodacom and MTN. Although not as prominent as MTN's, Vodacom shows a significant amount of fluctuation. MTN shares have significant price fluctuations, particularly in the middle of 2023 when there are noticeable price increases as shown in Fig. 2. This volatility may result from a number of factors, like changes in market mood, market reactions to earnings announcements, or external economic events. Telkom's share price is less erratic, fluctuating less in both directions. However, there is a discernible fall in the middle of the year, which may indicate a challenging period for the company or issues specific to the industry. Though the difference in size is less, Vodacom has a decline in the middle of 2023, much like MTN. MTN's share price is still relatively high compared to the others despite the volatility; nevertheless, there is a tendency for the price to decline by the end of 2023, which could indicate that the company is experiencing unfavorable market conditions or negative investor sentiment. By the end of 2023, Telkom's share price looks to be somewhat rebounding, indicating some improved market conditions. Like Telkom, Vodacom's share price shows a slight increase towards the end of the year, indicating either future stability or positive improvements in the market for the company.

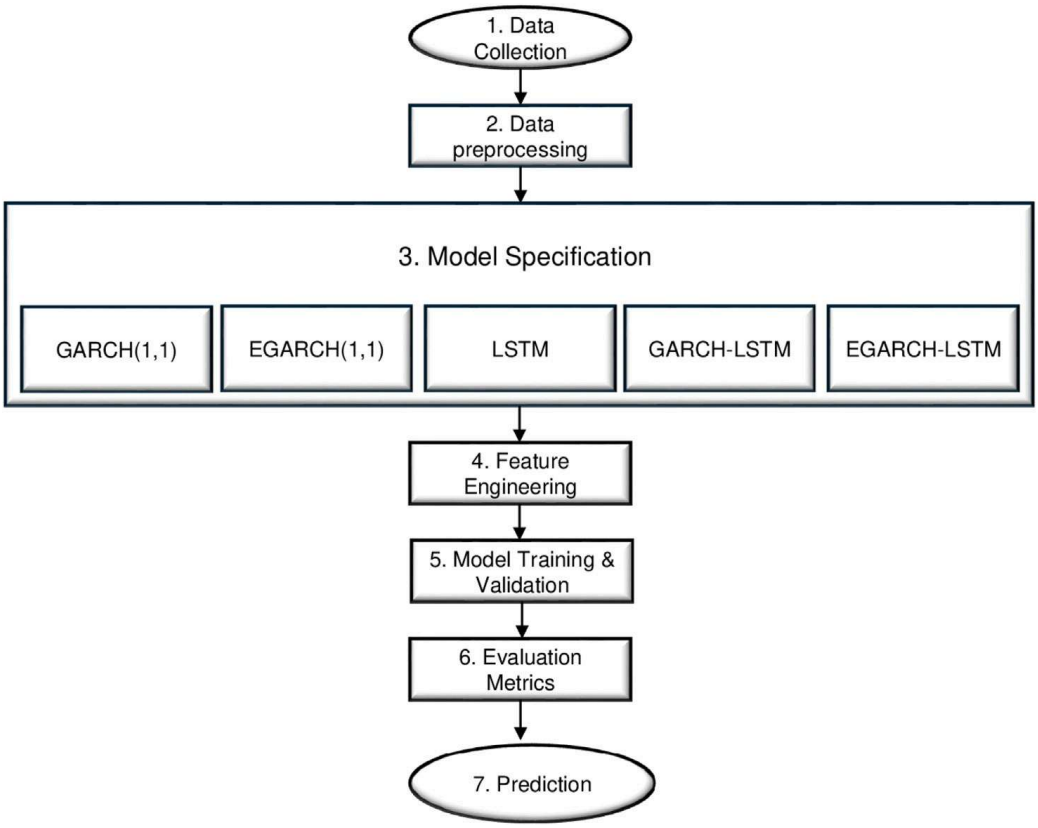

Fig. 1. Comprehensive hybrid approach to volatility prediction, integrating GARCH, EGARCH and LSTM techniques.

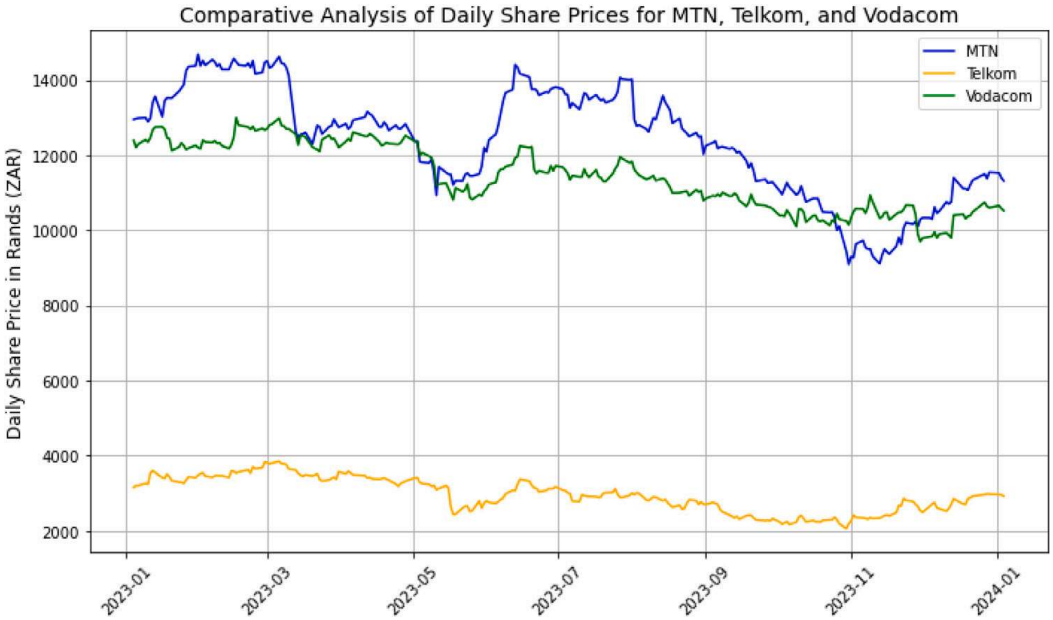

Fig. 2. The plot provides a visual comparison of the daily share prices of three major South African telecommunication companies: MTN, Telkom, and Vodacom over the period from early 2023 to early 2024.

Data preprocessing

Table 2 presents descriptive statistics for MTN, Telkom, and Vodacom stocks over 250 trading days, with prices denominated in South African Rand (ZAR). The table reports the number of observations Count, Mean, Min and Max prices, Std, and trading volumes. MTN has the highest average stock prices across all categories (e.g., an average opening price of R12,373.21) and the

Table 2
Summary statistics for MTN, Telkom, and Vodacom stocks.

Stock	Statistic	Open	High	Low	Close	Adj close	Volume
MTN	Count	250	250	250	250	250	250
	Mean	12,373.21	12,550.62	12,193.35	12,375.67	12,286.81	4,795,777
	Min	9,075.00	9,341.00	8,700.00	9,098.00	9,098.00	178,176
	Max	14,687.00	14,942.00	14,429.00	14,687.00	14,410.00	23,645,010
	Std	1,450.85	1,472.87	1,455.15	1,467.70	1,389.39	2,964,200
Telkom	Count	250	250	250	250	250	250
	Mean	2,947.35	3,014.88	2,885.74	2,949.23	2,949.23	1,406,904
	Min	2,055.00	2,228.00	2,040.00	2,060.00	2,060.00	30,638
	Max	3,835.00	3,965.00	3,784.00	3,850.00	3,850.00	9,191,587
	Std	449.78	458.23	451.14	454.82	454.82	1,099,650
Vodacom	Count	250	250	250	250	250	250
	Mean	11,460.94	11,565.58	11,298.40	11,431.92	10,979.60	1,416,693
	Min	9,602.00	9,831.00	9,617.00	9,700.00	9,700.00	71,720
	Max	13,199.00	13,746.00	12,776.00	13,002.00	12,280.97	8,557,127
	Std	875.81	873.96	861.52	871.42	689.98	956,311

largest average daily trading volume (approximately 4.8 million shares), indicating strong market activity. Telkom has the lowest average prices (around R2,947.35) and lowest minimum prices, reflecting its smaller market capitalization. Vodacom's average opening price (R11,460.94) is lower than MTN's but higher than Telkom's, and its adjusted closing price (R10,979.60) is intermediate between MTN and Telkom. In terms of volatility, MTN exhibits the highest standard deviation across price measures, indicating the greatest price fluctuations, while Telkom shows the lowest standard deviations. Vodacom's price variability falls between MTN and Telkom, suggesting moderate volatility. These statistics provide insight into price levels, fluctuations, and trading activity, which are essential for assessing the risk and potential returns of each stock. This analysis helps investors understand the risk and potential return associated with these stocks. To calculate the daily logarithmic return (continuously compounded return) [36], we used the following formula:

$$r_t = \log\left(\frac{P_t}{P_{t-1}}\right), \tag{1}$$

where r_t is the daily return at time t, P_t is the adjusted closing price at time t, and P_{t-1} is the adjusted closing price at the previous time $t-1$. As shown in Eq. (1), the return represents the percentage change between consecutive closing prices, which is illustrated in Fig. 3. The graphs visually demonstrates how the daily returns of each stock fluctuate, providing insights into the volatility and risk associated with these companies over time. The MTN daily returns are quite volatile, with several large upward and negative surges. This suggests that MTN's stock is more volatile on a daily basis, which reflects more risk. The high volatility is in line with the share price plot, which showed more significant price fluctuations for MTN. Telkom's daily returns exhibit even more marked volatility, with many sizable spikes, many of which are positive indicating that the company has periods of notable daily gains. But there are also notable negative returns, which suggests a high level of risk. In comparison to MTN and Telkom, Vodacom's daily returns exhibit more stability and less swings. This implies that, among the three telecommunication firms operating during this time, Vodacom may be the least volatile and, hence, less hazardous investment. Based on the daily returns, Telkom appears to be the most volatile and therefore carries the highest risk. MTN also exhibits high volatility but to a slightly lesser extent than Telkom. Vodacom, while not immune to volatility, appears to have the most stable returns, which might indicate a lower risk profile. Reflecting the overall ups and downs in the market, all three equities exhibit periods of both positive and negative returns. Telkom may be more unpredictable because to its increased susceptibility to market news and events, as seen by the greater frequency and amplitude of dramatic returns. All three equitie's returns cluster around zero, indicating that most days see modest returns that are indicative of a generally steady market. The spikes, however, imply that there may occasionally be announcements or events that result in notable changes in price. Several causes might be responsible for the observed volatility, such as stock prices can be greatly impacted by news of new company endeavors, leadership changes, or earnings releases. Market-wide volatility can be brought on by broader economic factors including shifts in interest rates, the publication of economic data, or developments in geopolitics. Sharp swings in stock prices can result from shifts in investor attitude, which are frequently caused by market speculation.

Fig. 4 shows the distribution of returns for MTN, Telkom, and Vodacom. The MTN returns distribution appears slightly skewed to the left (negatively skewed). The peak of the histogram is centered slightly above 0, indicating that the most common returns are near zero, with a slight tendency towards positive values. The returns generally range between approximately −0.10 and 0.075, with most clustering around 0. The Telkom returns distribution is also slightly skewed to the left but appears more symmetric compared to MTN. The returns are centered around 0, with a slight concentration of higher frequencies near this point. The returns range from approximately −0.15 to 0.10, indicating greater volatility compared to the other stocks. The Vodacom returns distribution is approximately symmetric and bell-shaped, resembling a normal distribution. The histogram is centered close to 0, showing that the most frequent returns are around zero. The returns range between approximately −0.04 and 0.06, exhibiting a narrower spread compared to MTN and Telkom, which suggests lower volatility. Overall, Telkom exhibits the highest volatility with the greatest spread, followed by MTN, while Vodacom has the lowest volatility. While MTN and Telkom show minor negative skewness, Vodacom's returns are more symmetrical and closely resemble a normal distribution. The returns for all three equities are primarily concentrated around zero but differ in terms of outlier prevalence and dispersion, as shown in Fig. 4.

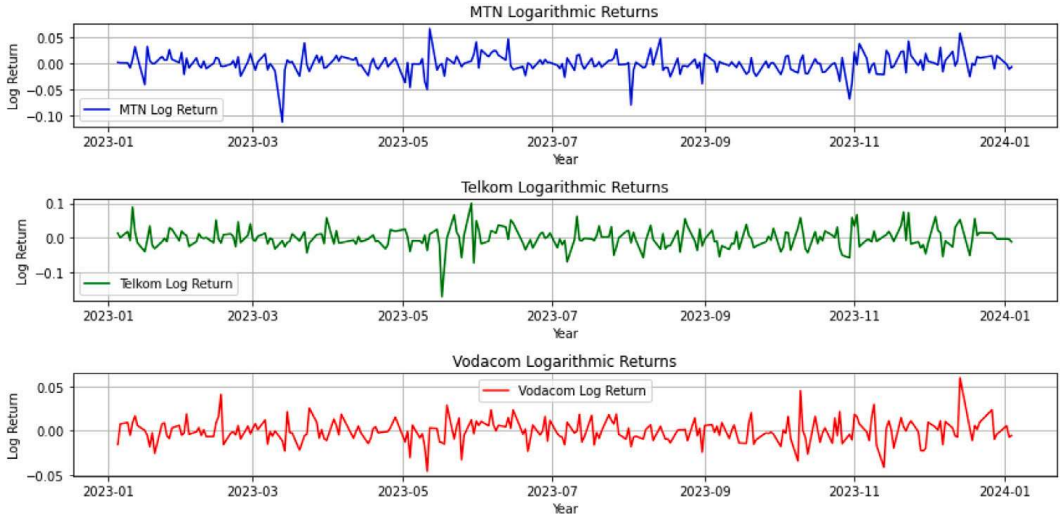

Fig. 3. The plot shows the daily returns for MTN, Telkom, and Vodacom over the same period as the previous share price comparison. Daily returns are a measure of the day-to-day percentage change in the share price, which gives insight into the volatility and risk associated with each stock.

Model specification

ARCH model

Financial time series, such as stock returns, often exhibit periods of high volatility followed by periods of relative calm, a phenomenon known as volatility clustering. To capture this behavior, Engle [2] introduced the ARCH model. The basic idea behind ARCH is that today's market volatility may depend on the size of past market shocks. For instance, if the market was surprised yesterday, we might expect higher uncertainty (volatility) today. The return series r_t is modeled as:

$$r_t = \mu + \epsilon_t, \tag{2}$$

where μ is the average return, and ϵ_t is the unpredictable component or shock. This shock is further modeled as:

$$\epsilon_t = \sigma_t z_t, \tag{3}$$

where $z_t \sim N(0,1)$ is standard white noise, and σ_t is the time-varying standard deviation our measure of volatility. The ARCH(q) model specifies the conditional variance as a function of past squared shocks:

$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^{q} \alpha_i \epsilon_{t-i}^2, \tag{4}$$

where $\alpha_0 > 0$ ensures positive variance, and $\alpha_i \geq 0$ ensures non-negativity of the volatility estimate. If large shocks occurred recently (i.e., large ϵ_{t-i}^2), the model expects higher volatility today σ_t^2. This allows ARCH to capture real-world phenomena like market panic or excitement spreading over time.

GARCH model

While ARCH models account for volatility clustering by linking current volatility to past squared returns, they are limited in how much "memory" they have. To overcome this, Bollerslev [3] proposed the GARCH model, which adds greater flexibility by incorporating past volatility itself into the prediction. The GARCH model assumes that today's market uncertainty (volatility) depends not only on recent shocks (large price changes), but also on how volatile the market was in the recent past. This better reflects the real world, where markets often remain turbulent (or calm) over several days. The conditional variance in the GARCH(p, q) model is defined as:

$$\sigma_t^2 = \omega + \sum_{i=1}^{q} \alpha_i \epsilon_{t-i}^2 + \sum_{j=1}^{p} \beta_j \sigma_{t-j}^2, \tag{5}$$

where $\omega > 0$ is the baseline variance (long-run average), α_i is the sensitivity to past squared shocks (ARCH terms), β_j: sensitivity to past volatility (GARCH terms), ϵ_{t-i} is the past residuals (shocks), σ_{t-j}^2 is the past conditional variances. The most commonly used form is GARCH(1,1), which simplifies to:

$$\sigma_t^2 = \omega + \alpha_1 \epsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2. \tag{6}$$

For the variance to remain stable over time (i.e., not explode), we require:

$$\alpha_1 + \beta_1 < 1. \tag{7}$$

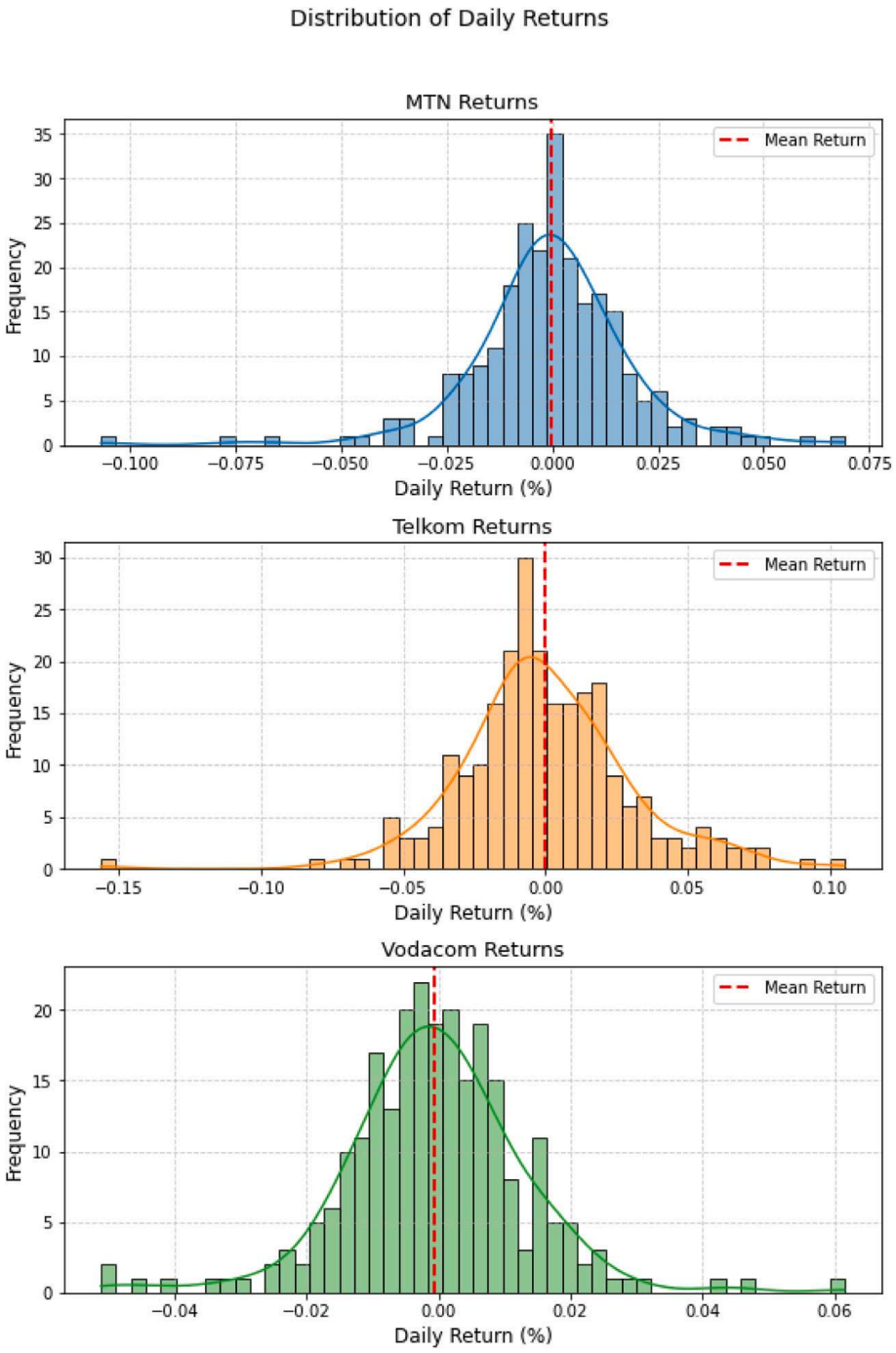

Fig. 4. This plot displays the distribution of daily returns for three telecommunication stocks: MTN, Telkom, and Vodacom.

Parameters $\theta = (\omega, \alpha_1, \beta_1)$ are estimated via Maximum Likelihood Estimation (MLE), using the following log-likelihood function:

$$\log L(\theta) = -\frac{n}{2}\log(2\pi) - \frac{1}{2}\sum_{t=1}^{n}\left(\log\sigma_t^2 + \frac{\epsilon_t^2}{\sigma_t^2}\right), \tag{8}$$

where n is the number of observations and $\epsilon_t = y_t - \mu$ is the residual. To evaluate the significance of parameter estimates, we use a Z-statistic:

$$Z_i = \frac{\hat{\theta}_i - \theta_{i,0}}{\mathrm{SE}(\hat{\theta}_i)}, \tag{9}$$

where $\hat{\theta}_i$ is the estimated parameter, and $\mathrm{SE}(\hat{\theta}_i)$ is its standard error. The p-value is then obtained by comparing Z_i to the standard normal distribution [37]. A low p-value (typically < 0.05) indicates significance. To compare different models, we use the Akaike

Information Criterion (AIC) [38] and the Bayesian Information Criterion (BIC) [39]. These criteria help balance the model's goodness-of-fit with its complexity. The AIC is defined as:

$$\text{AIC} = -2L(\hat{\theta}) + 2k, \tag{10}$$

where k is the number of estimated parameters and $L(\hat{\theta})$ is the log-likelihood evaluated at the estimated parameters. The BIC is defined as:

$$\text{BIC} = -2L(\hat{\theta}) + k\log(n), \tag{11}$$

where n is the sample size, k is the number of estimated parameters, and $L(\hat{\theta})$ is the log-likelihood evaluated at the estimated parameters. Smaller AIC and BIC values suggest a better model fit. AIC penalizes the number of parameters less than BIC, so BIC generally favors simpler models when comparing multiple models with the same data. GARCH models are highly effective for modeling and forecasting financial volatility, especially in markets that exhibit persistent periods of turbulence or calm.

EGARCH model

While GARCH models assume that good news and bad news affect volatility equally, this is often not the case in real markets. For example, a sudden drop in stock price (bad news) usually causes a sharper increase in volatility than a rise in price (good news) of the same magnitude. This phenomenon is known as the leverage effect or asymmetric volatility. To model this behavior, Nelson [5] introduced the Exponential GARCH (EGARCH) model, which allows volatility to respond differently to positive and negative shocks. Unlike standard GARCH models that model the variance directly, EGARCH models the logarithm of the conditional variance. This ensures that the variance is always positive without needing to impose non-negativity constraints on the parameters. The general EGARCH(p, q, r) model is given by:

$$\log(\sigma_t^2) = \omega + \sum_{j=1}^{q} \beta_j \log(\sigma_{t-j}^2) + \sum_{i=1}^{p} \alpha_i \left| \frac{\epsilon_{t-i}}{\sigma_{t-i}} \right| + \sum_{k=1}^{r} \gamma_k \frac{\epsilon_{t-k}}{\sigma_{t-k}}, \tag{12}$$

where σ_t^2 is the conditional variance at time t (volatility), ϵ_t is the residual or shock at time t, $\frac{\epsilon_t}{\sigma_t}$ is the standardized shock, α_i captures the magnitude effect (how big the shock was), γ_k captures the asymmetry (direction of the shock), β_j captures volatility persistence (memory of past volatility). The inclusion of both absolute and signed terms allows the EGARCH model to capture not only how large a shock was but also whether it was positive or negative. The most commonly used version is EGARCH(1,1), which simplifies to:

$$\log(\sigma_t^2) = \omega + \beta_1 \log(\sigma_{t-1}^2) + \alpha_1 \left| \frac{\epsilon_{t-1}}{\sigma_{t-1}} \right| + \gamma_1 \frac{\epsilon_{t-1}}{\sigma_{t-1}}, \tag{13}$$

The term $\left|\frac{\epsilon_{t-1}}{\sigma_{t-1}}\right|$ increases volatility in response to large shocks, regardless of direction. The term $\frac{\epsilon_{t-1}}{\sigma_{t-1}}$ allows negative shocks to have a different (typically larger) effect than positive ones. EGARCH is a powerful model for financial volatility forecasting because it naturally handles asymmetry and guarantees positive variance. It is particularly useful in markets where bad news causes stronger reactions than good news.

LSTM

LSTM is a variant of recurrent neural networks (RNNs) tailored for sequence prediction tasks such as time series forecasting. LSTMs effectively capture long-term dependencies in data by maintaining a memory of past information. The core equations for an LSTM cell are:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \tag{14}$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \tag{15}$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \tag{16}$$

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t \tag{17}$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \tag{18}$$

$$h_t = o_t \cdot \tanh(C_t). \tag{19}$$

The LSTM cell operates through a series of gates and activations to process sequential data. At each time step t, the forget gate f_t determines how much of the previous cell state C_{t-1} should be retained, based on the input at time t (x_t) and the previous hidden state h_{t-1}. The input gate i_t controls how much new information should be added to the cell state, while the candidate cell state $\tilde{C}_t$ represents potential new memory content. The cell state C_t is updated by combining the forget gate's contribution (forgetting part of the previous memory) with the input gate's contribution (adding new memory). The output gate o_t determines what part of the cell state should be output as the hidden state h_t, which serves as both the output of the LSTM at time t and as input to the next time step. Each gate has associated weight matrices W_f, W_i, W_C, W_o and bias terms b_f, b_i, b_C, b_o that are learned during training to optimize the model's ability to capture long-term dependencies in the data. LSTM models are powerful for time series prediction as they can preserve crucial information across long sequences, making them ideal for capturing temporal dependencies in financial data.

Proposed hybrid models with attention mechanism

In this section, we introduce a novel hybrid model that integrates the GARCH model, as discussed in Section "GARCH model", the EGARCH model in Section "EGARCH model", and the LSTM model in Section "LSTM" with an attention mechanism. This integration aims to improve prediction accuracy by enabling the model to focus on critical time steps. The hybrid model leverages the strengths of volatility forecasting, sequence learning, and dynamic weighting of significant time periods. To enhance the predictive performance of the GARCH-LSTM and EGARCH-LSTM hybrid models, an attention mechanism is incorporated to dynamically focus on the most relevant time steps in the sequence. Given the LSTM hidden states h_t at each time step, we compute an attention score to assess their relative importance:

$$\text{score}(h_t) = v^T \tanh(W_h h_t + b_h), \tag{20}$$

where W_h and b_h are trainable weight and bias parameters, v is a learnable vector that helps determine the importance of h_t, and $\tanh(\cdot)$ is a non-linear activation function. The scores are converted into normalized attention weights α_t using the softmax function:

$$\alpha_t = \frac{\exp(\text{score}(h_t))}{\sum_{k=1}^{T} \exp(\text{score}(h_k))}, \tag{21}$$

where $\sum_{t=1}^{T} \alpha_t = 1$, ensuring a proper probability distribution, and T is the total number of time steps. The context vector $\hat{h}$ represents a weighted sum of the LSTM hidden states:

$$\hat{h} = \sum_{t=1}^{T} \alpha_t \cdot h_t \tag{22}$$

This step effectively emphasizes important time steps while suppressing less relevant information. The final forecast $\hat{y}_t$ is obtained by passing the context vector and the volatility forecast σ_t^2 (from GARCH or EGARCH) through a fully connected layer:

$$\hat{y}_t = f_{\text{FC}}(\hat{h}, \sigma_t^2), \tag{23}$$

where f_{FC} is a function that merges the attention-enhanced sequence features with volatility estimates to make the final prediction. The only difference between the GARCH-LSTM-Attention and EGARCH-LSTM-Attention models lies in the volatility modeling step: the former uses the symmetric GARCH(1,1) process, while the latter employs EGARCH(1,1) to capture asymmetries in volatility. In both models, the LSTM and attention mechanisms are structurally and functionally identical, as they both utilize the LSTM network to model temporal dependencies. Similarly, the attention mechanism including the computation of attention scores, softmax normalization, and context vector generation is exactly the same across both architectures. Finally, the prediction step in both models involves concatenating the context vector $\hat{h}$ with the estimated conditional variance σ_t^2 and passing this through a fully connected layer to produce the forecast, which is also identical between the two models. This architectural consistency allows us to isolate and evaluate the impact of the volatility estimator on overall forecasting performance. The complete forecasting pipeline is summarized in Algorithm 1.

Algorithm 1: Proposed Hybrid GARCH/EGARCH-LSTM-Attention Forecasting Model

Input: Stock returns R_t, technical indicators I_t (e.g., SMA, EMA, RSI), model type (GARCH or EGARCH)
Output: Forecasted return or volatility $\hat{y}_{t+1}$

Step 1: Volatility Estimation
if *model type is GARCH* **then**
  Estimate σ_t^2 using GARCH(1,1)
else if *model type is EGARCH* **then**
  Estimate σ_t^2 using EGARCH(1,1)
Step 2: Input Construction
Construct sequence $X_t = [R_t, \sigma_t^2, I_t]$ **Step 3: Temporal Modeling with LSTM**
Pass X_t through LSTM to obtain hidden states $H = [h_1, h_2, \ldots, h_T]$ **Step 4: Attention Score Computation**
for $t \leftarrow 1$ **to** T **do**
  $\text{score}_t \leftarrow v^\top \tanh(W_h h_t + b_h)$
Step 5: Attention Weight Normalization
$Z \leftarrow \sum_{k=1}^{T} \exp(\text{score}_k)$
for $t \leftarrow 1$ **to** T **do**
  $\alpha_t \leftarrow \frac{\exp(\text{score}_t)}{Z}$
Step 6: Context Vector Calculation
$\hat{h} \leftarrow \sum_{t=1}^{T} \alpha_t \cdot h_t$ **Step 7: Final Forecast**
Concatenate $\hat{h}$ with latest volatility estimate σ_t^2
$\hat{y}_{t+1} \leftarrow f_{\text{FC}}(\hat{h}, \sigma_t^2)$ **return** $\hat{y}_{t+1}$

Let n denotes the number of time steps in the input sequence and h the number of hidden units in the LSTM layer. The computational complexity is as follows:

- Volatility estimation (GARCH/EGARCH): $\mathcal{O}(n)$
- LSTM forward pass: $\mathcal{O}(nh^2)$
- Attention mechanism: $\mathcal{O}(n^2)$

The total computational complexity is $\mathcal{O}(n^2)$, with the attention mechanism being the dominant factor. However, this remains acceptable for typical financial time series of moderate length (e.g., 250–300 daily observations). The model's hyperparameters were tuned using a random grid search method combined with rolling validation (see Table 3). All settings were selected to balance computational efficiency and forecasting accuracy, using MAE, RMSE, and MSE as evaluation metrics.

Feature engineering

In the domain of time series forecasting, especially for financial market prediction, feature engineering is a critical step. In this work, we focus on using technical indicators to enhance the predictive capability of GARCH-LSTM and EGARCH-LSTM models. The technical indicators are SMA, EMA and RSI. The SMA is calculated as the average of the closing prices over a specified number of periods, denoted as n:

$$SMA_t = \frac{1}{n}\sum_{i=0}^{n-1} P_{t-i}, \tag{24}$$

where SMA_t is the Simple Moving Average at time t, P_{t-i} is the price at time $t-i$, n is the number of periods. The SMA helps reduce price noise and identifies overall trends. The EMA gives greater weight to more recent prices, making it more responsive to the latest market movements. It is calculated recursively as:

$$EMA_t = \alpha P_t + (1-\alpha)EMA_{t-1}, \tag{25}$$

where EMA_t is the Exponential Moving Average at time t, P_t is the price at time t, $\alpha = \frac{2}{n+1}$ is the smoothing factor, with n representing the number of periods. The EMA is more sensitive to recent price changes, providing a better reflection of current market trends. The RSI evaluates the velocity and extent of price movements to detect overbought or oversold market conditions. It is calculated as:

$$RSI_t = 100 - \frac{100}{1+RS_t}, \tag{26}$$

where RS_t = average gain/average loss over a given period (commonly 14 days). The RSI values span from 0 to 100, where levels above 70 signal overbought conditions, while those below 30 indicate oversold conditions. The analysis of MTN in Fig. 5(a) and Telkom in Fig. 5(b) stock prices in 2023 reveals similar trends both peaked around March-April, declined mid-year, and showed signs of recovery from November onward. The SMA and EMA confirm these trends, with EMA responding more quickly to price changes. The RSI highlights potential trading signals, indicating overbought conditions (above 70) in early and late 2023 and oversold conditions (below 30) around May-June. MTN exhibits higher volatility compared to Telkom, with sharper fluctuations. Overall, these indicators suggest that both stocks followed cyclical trends, with clear buying and selling opportunities based on RSI levels. The stock price peaked in early 2023 before declining steadily from mid-year, with SMA and EMA confirming the downward trend as shown in Fig. 5(c) for Vodacom. The EMA reacts more quickly to price movements, aligning closely with the price action. The RSI indicates overbought conditions (above 70) early in the year and towards the end, suggesting potential corrections, while it drops below 30 around June, signaling oversold conditions and a possible buying opportunity. Overall, Vodacom's stock followed a declining trend for most of the year, with short-term rebounds near the end.

Model training & Validation

Table 3 shows the hyperparameters used. To forecast stock prices, we combine traditional econometric models, DL techniques, and hybrid models that integrate both approaches. We utilize GARCH and EGARCH models to capture the changing volatility in stock returns, helping to understand patterns like volatility clustering. In addition, LSTM networks are employed to identify long-term dependencies in stock price movements. To enhance performance, we combine GARCH/EGARCH models with LSTM with attention mechanism networks in hybrid architectures, known as GARCH-LSTM with attention and EGARCH-LSTM with attention. These hybrid models take advantage of both volatility modeling and DL, with attention mechanisms added to the LSTM networks to focus on critical time steps, enhancing the model's capability to generate precise predictions. The models are fine-tuned using random grid search, which explores various hyperparameters such as the number of LSTM units, layers, dropout rates, learning rates, batch sizes, and epochs. This approach ensures that the models are optimized for the best possible performance. We train and validate the models using stock returns, technical indicators, and volatility forecasts, with the dataset split into 80% training and 20% testing. Performance is assessed using metrics like RMSE, MAE, and MSE, which help determine how well the models predict stock prices and ensure their accuracy in forecasting. The training and testing process was conducted on a Dell latest-generation system running Windows 11, equipped with 16 GB RAM to efficiently handle DL computations. The software environment included Python 3.11.6, TensorFlow 2.15.0 for DL, TA-Lib 0.4.0 for technical analysis, Pandas 2.2.0 for data manipulation, NumPy 1.26.3 for numerical operations, ARCH 5.7.0 for GARCH/EGARCH volatility modeling, and Statsmodels 0.14.1 for statistical analysis.

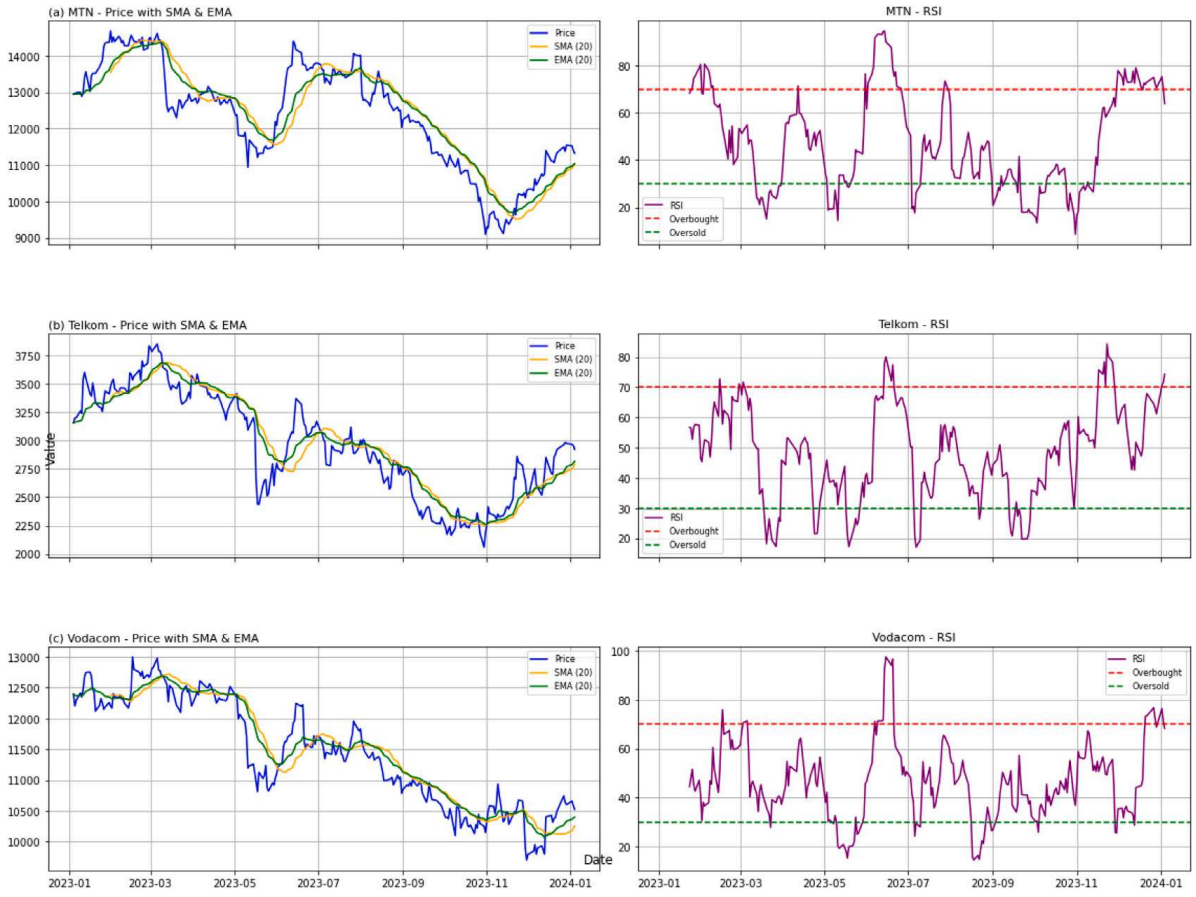

Fig. 5. Stock price with SMA, EMA, and RSI for (a) MTN, (b) Telkom, and (c) Vodacom.

Table 3
Hyperparameter tuning for hybrid model (GARCH/EGARCH + LSTM with attention) using grid search.

Hyperparameter	Explored values	Best value
GARCH/EGARCH distribution	{normal, student-t, skew-t}	student-t
Rolling Window Size (Log returns & Technical indicators)	{10, 20, 30, 40}	20
LSTM units	{50, 100, 150}	100
Number of LSTM layers	{1, 2, 3}	2
Dropout rate	{0.2, 0.3, 0.5}	0.3
Hidden layer size (Dense layers)	{32, 64, 128}	64
Optimizer	{Adam, RMSprop}	Adam
Learning rate	{0.001, 0.0005, 0.0001}	0.0001
Batch size	{16, 32, 64}	32
Epochs	{10, 20, 30}	20
Early stopping (Patience)	{10, 20, 30}	20
Loss function	{MSE, MAE, Huber}	MSE
Attention activation function	{tanh, relu, softmax}	tanh
Attention layer units	{16, 32, 64}	32

Evaluation metrics

In this work, several evaluation metrics were used to assess the performance of the volatility forecasting models: MAE, MSE and RMSE. These metrics allow for the assessment of prediction accuracy and provide insight into how well the models fit the data. Where by y_i is the actual value, $\hat{y}_i$ is the predicted value, and n is the number of data points.

The MAE measures the average magnitude of errors in the predictions, irrespective of their direction. It is computed as:

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|. \tag{27}$$

Table 4
GARCH(1,1) results and diagnostic tests for MTN, Telkom, and Vodacom.

Metric/Parameter	GARCH(1,1) model results			Diagnostic tests		
	MTN	Telkom	Vodacom	MTN	Telkom	Vodacom
Mean (μ)	−0.00031 (p = 0.795)	−0.00026 (p = 0.883)	−0.00064 (p = 0.632)	–	–	–
Omega (ω)	0.00034 (p = 0.046)	0.00027 (p = 0.161)	0.0000038 (p = 0.000)	–	–	–
Alpha (α_1)	0.0731 (p = 0.605)	0.0928 (p = 0.107)	0.0100 (p = 0.993)	–	–	–
Beta (β_1)	3.16e−13 (p = 1.000)	0.6070 (p = 0.006)	0.9700 (p = 0.444)	–	–	–
Log-Likelihood	631.945	521.881	714.510	–	–	–
AIC	−1255.89	−1035.76	−1421.02	–	–	–
BIC	−1241.82	−1021.69	−1406.95	–	–	–
Number of observations	249	249	249	–	–	–
ARCH-LM test p-value	–	–	–	0.9978	0.1933	0.5201
Jarque–Bera test p-value	–	–	–	9.16×10^{-97}	7.56×10^{-37}	1.93×10^{-20}

The MSE measures the average of the squares of the errors. It penalizes large errors more heavily than MAE, which can be useful in identifying models that make large mistakes. It is calculated as:

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2. \tag{28}$$

The RMSE is the square root of MSE and has the same units as the target variable, making it more interpretable. It is sensitive to large errors and thus emphasizes larger deviations between predicted and actual values. It is defined as:

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}. \tag{29}$$

These evaluation metrics are critical for understanding the quality of the models used in this study. By applying them, we can quantitatively assess how well each model is performing in forecasting stock volatility and ultimately make informed decisions regarding the most suitable model for predicting market behavior.

Results and discussion

In this section, we present the estimation results for the GARCH(1,1) and EGARCH(1,1) models, along with their respective diagnostic tests. Additionally, we include the results of the hybrid models, GARCH-LSTM and EGARCH-LSTM, both with and without the attention mechanism. These models are used to predict volatility and assess the robustness of volatility forecasting in the context of South African telecommunication equities (MTN, Telkom, and Vodacom).

GARCH estimation

The results of the GARCH(1,1) model applied to the stock returns of MTN, Telkom, and Vodacom are summarized in Table 4 with the diagnostic test. The key parameters omega (ω), alpha (α_1), and beta (β_1) along with their associated p-values are presented for each stock.

This analysis of the GARCH(1,1) model results offers valuable insights into the behavior of volatility dynamics of MTN, Telkom, and Vodacom. For MTN, the significant ω suggests that long-term average volatility plays a crucial role, while the insignificance of both α_1 and β_1 indicates that recent shocks and volatility persistence have minimal impact, implying relatively stable volatility. In contrast, Telkom's ω is not significant, but its β_1 is, showing strong volatility persistence—meaning that once volatility rises, it tends to remain high. The near-significant β_1 suggests that short-term shocks may also play a role. Vodacom's model highlights a highly significant ω, indicating that long-term variance is a key driver of its volatility, but the insignificance of α_1 implies that short-term shocks and volatility persistence are not substantial factors. Notably, Vodacom's model has the lowest AIC, suggesting it might be the best-fitting model among the three. In summary, MTN and Vodacom exhibit stable long-term volatility with little influence from recent shocks, while Telkom demonstrates stronger volatility persistence.

Model diagnostics

After fitting the GARCH model, diagnostic tests were conducted to check for any remaining heteroscedasticity and ensure the model adequately explains the variance in the returns. The following tests were performed: the ARCH-LM [2] test to detect any remaining ARCH effects, and the Jarque–Bera [40] test for normality of the residuals. The findings of these tests are shown in Table 4. The ARCH-LM test results show no significant remaining ARCH effects (p-value > 0.05), indicating that the model adequately captures volatility clustering. The Jarque–Bera test findings suggest that the residuals are not normally distributed. The residuals plot serves as a visual confirmation of the absence of discernible patterns or structures within the residuals, which constitutes a crucial assessment for the adequacy of model fitting. This observation complements the findings of the ARCH-LM and Jarque–Bera tests, which indicate that the model sufficiently captures the phenomenon of volatility clustering. The Autocorrelation Function (ACF) plot facilitates the identification of any significant autocorrelation that may persist in the residuals, thereby suggesting that the model

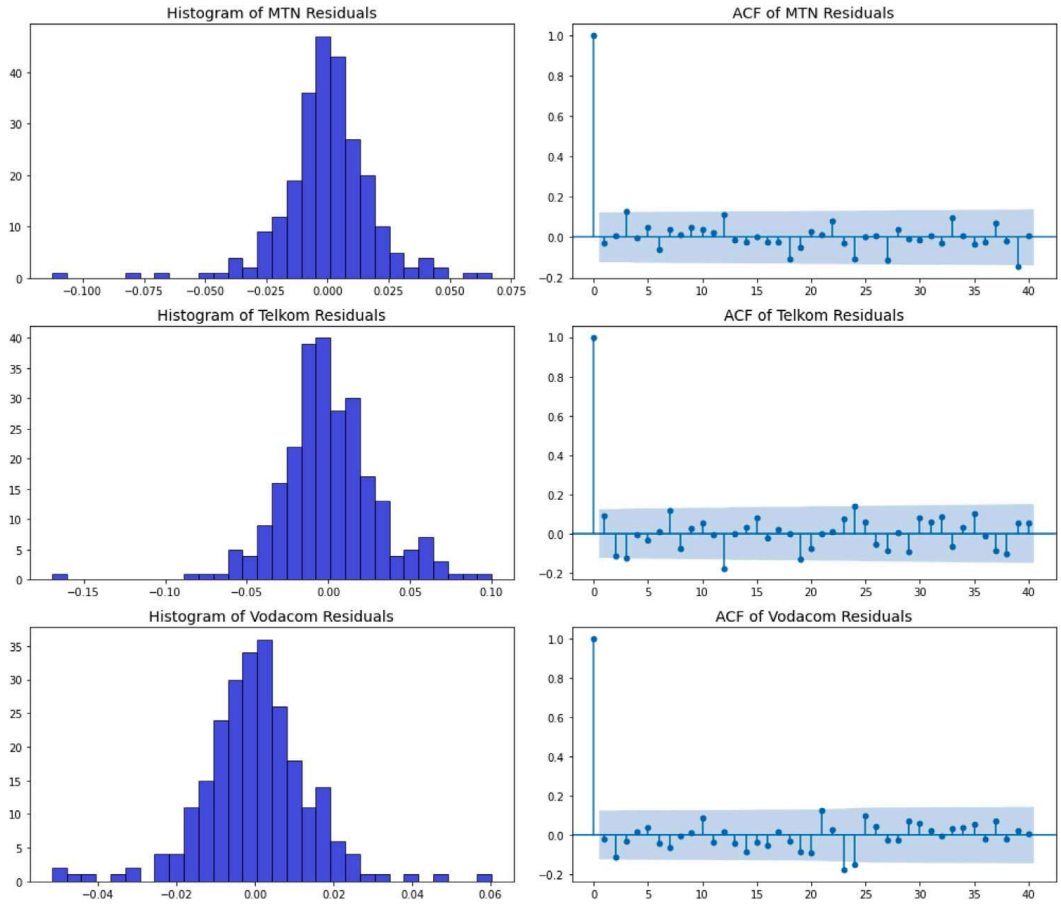

Fig. 6. Histograms and ACF Plots of Residuals for MTN, Telkom, and Vodacom Stocks generated by GARCH. The histograms (left column) suggest that the residuals are approximately normally distributed, while the ACF plots (right column) indicate minimal autocorrelation, implying that the fitted models have effectively captured the time dependencies in the data.

may not have entirely accounted for the underlying dynamics of the data. The incorporation of the ACF plot Fig. 6 further bolsters the assessment of model adequacy. Moreover, the Jarque–Bera test highlights the non-normality of the residuals, a condition that can be further elucidated through a histogram representing the residuals. This histogram offers a visual depiction of the deviation from normality, underscoring the necessity for potential modifications or additional considerations in the refinement of the model. The histogram of residuals for MTN as shown in Fig. 6 indicates that the distribution is not perfectly normal, which is further confirmed by the very small p-value from the Jarque–Bera test, suggesting significant deviation from normality. Despite the GARCH model effectively capturing volatility clustering, the non-normality of the residuals points to the potential need for further model adjustments, possibly considering non-Gaussian distributions. The ACF plot of the residuals shows no significant autocorrelation, suggesting that the model has successfully captured the time-dependent structure in the dataset. This is also validated by the ARCH-LM test, which shows no significant remaining ARCH effects (p-value > 0.05), suggesting that the model has adequately accounted for volatility clustering. The histogram of residuals for Telkom as shown in Fig. 6 indicates a departure from normality, as supported by the Jarque–Bera test (p-value $= 7.56 \times 10^{-37}$), suggesting significant deviation from a normal distribution. This non-normality suggests that the model may need further modifications, such as integrating alternative distributions for the residuals or enhancing the model's flexibility to better account for fat tails or skewness in the return distribution. The ACF plot of Telkom's residuals shows no significant autocorrelations, implying that the model has successfully captured the autocorrelations present in the dataset. This observation is consistent with the ARCH-LM test result, which indicates no significant remaining ARCH effects, indicating that the model has sufficiently captured the volatility clustering. The histogram of residuals for Vodacom as shown in Fig. 6 reveals a significant departure from normality, as indicated by the Jarque–Bera test p-value, suggesting a substantial deviation from a normal distribution. This non-normality points to the potential presence of fat tails or skewness within the residuals, which the GARCH model did not fully capture. To address this, further investigation into alternative distributions, such as a student's t-distribution or skewed distributions, may be necessary to better model the dataset's characteristics. The ACF plot of Vodacom's residuals shows no significant autocorrelation, indicating that the model has effectively captured the time-series dependencies in the data. This observation is corroborated by the ARCH-LM test findings, which demonstrate no significant remaining ARCH effects, validating that the GARCH model has sufficiently addressed volatility clustering.

Table 5
EGARCH model results and diagnostic test results for MTN, Telkom, and Vodacom.

Parameter/Metric	EGARCH model results			Diagnostic test results		
	MTN	Telkom	Vodacom	MTN	Telkom	Vodacom
Mean (μ)	−0.00036 (p = 0.759)	−0.00008 (p = 0.967)	−0.00052 (p = 0.776)	–	–	–
Omega (ω)	−7.1575 (p = 0.003)	−2.2904 (p = 0.020)	−0.1934 (p = 0.000)	–	–	–
Alpha (α_1)	0.1658 (p = 0.318)	0.2736 (p = 0.010)	−0.1282 (p = 0.000)	–	–	–
Beta (β_1)	0.0952 (p = 0.765)	0.6723 (p = 0.000)	0.9787 (p = 0.000)	–	–	–
Log-Likelihood	634.220	522.946	724.465	–	–	–
AIC	−1260.44	−1037.89	−1440.93	–	–	–
BIC	−1246.37	−1023.82	−1426.86	–	–	–
Number of Obs.	249	249	249	–	–	–
Ljung–Box test p-value	–	–	–	0.6799	0.1179	0.8556
Jarque–Bera test p-value	–	–	–	0.0	0.0	2.89×10^{-15}
ADF test p-value	–	–	–	4.80×10^{-29}	4.64×10^{-5}	1.03×10^{-22}
ARCH LM test p-value	–	–	–	0.9482	0.5183	0.7347

Forecasted volatility for MTN, Telkom, and Vodacom

The forecasted volatility for the next 10 days was obtained using the GARCH(1,1) model for each of the three telecommunication stocks: MTN, Telkom, and Vodacom. The findings are presented in Table A.3. The forecasted volatility values for the three telecommunication stocks show distinct patterns. For MTN and Vodacom, the forecasted volatility remains relatively stable over the next 10 days, ranging from 1.84 to 1.91 for MTN and 1.60 to 1.62 for Vodacom. This indicates that these stocks are expected to experience consistent price fluctuations in the short term. In contrast, Telkom's forecasted volatility is higher and shows an increasing trend, ranging from 2.66 to 2.99 over the 10-day period. This suggests that Telkom is expected to experience larger price fluctuations compared to MTN and Vodacom. The persistent stability in MTN and Vodacom's volatility can be seen as a sign of less uncertainty in their price movements, whereas the rising volatility of Telkom indicates greater market uncertainty and risk.

EGARCH estimation

The findings derived from the EGARCH(1,1) model pertaining to MTN, Telkom, and Vodacom exhibit distinct characteristics in their respective volatility frameworks as shown in Table 5. In the case of MTN, the mean (μ) is established as negative; however, it lacks statistical significance (p = 0.759), which implies an absence of substantial drift in returns. The omega (ω) parameter is also negative and statistically significant (p = 0.003), thereby indicating the existence of persistent volatility. The alpha (α_1) coefficient for MTN is relatively low (0.1658), and the beta (β_1) coefficient is similarly modest, reflecting a moderate degree of volatility persistence. For Telkom, the mean (μ) is likewise negative yet statistically insignificant (p = 0.967), although the omega (ω) parameter is significant (p = 0.020), suggesting the presence of volatility persistence. The α coefficient (0.2736) is statistically significant (p = 0.010), indicating a responsive relationship of volatility to market shocks. The β value (0.6723) is comparatively high, suggesting substantial long-term volatility persistence. Vodacom presents a negative but statistically insignificant mean (μ) since (p = 0.776). The omega (ω) parameter is also negative and highly significant (p = 0.000), which indicates a pronounced presence of volatility within the series. The α coefficient is negative (−0.1282) and statistically significant (p = 0.000), potentially reflecting an inverse relationship between shocks and volatility, while the elevated β value (0.9787) signifies a very high degree of long-term volatility persistence.

Model diagnostics

Table 5 also shows the diagnostic test results for the EGARCH model applied to MTN, Telkom, and Vodacom stocks indicate that the model effectively captures key statistical properties of the data. The Ljung–Box [41] test p-values (MTN: 0.6799, Telkom: 0.1179, Vodacom: 0.8556) suggest no significant autocorrelation within the residuals, confirming that the model accounts for time dependencies. However, the Jarque–Bera [40] test p-values (0.0 for MTN and Telkom, 2.89×10^{-15} for Vodacom) indicate that the residuals are not normally distributed, likely due to skewness and excess kurtosis, which is typical in financial returns. The Augmented Dickey–Fuller (ADF) [42] test p-values (MTN: 4.80×10^{-29}, Telkom: 4.64×10^{-5}, Vodacom: 1.03×10^{-22}) confirm that the return series are stationary, making them suitable for time series modeling. Furthermore, the ARCH LM [2] test p-values (MTN: 0.9482, Telkom: 0.5183, Vodacom: 0.7347) suggest no significant heteroscedasticity within the residuals, implying that the EGARCH model has successfully captured volatility clustering. The volatility forecast for MTN shows a gradual increase over the next 10 days, starting at 0.0184 and rising slightly to 0.0192 by the end of the forecast horizon as shown in Table A.4. This indicates a relatively stable but slightly increasing volatility for MTN's stock returns over the short term. The volatility forecast for Telkom shows a more significant upward trend, starting at 0.0249 and reaching 0.0306 by day 10. This suggests increasing uncertainty or risk in Telkom's stock returns over the next 10 days, potentially reflecting market factors affecting the stock. Vodacom's volatility forecast is much lower than both MTN and Telkom, starting at 0.0115 and remaining relatively stable, with only slight fluctuations. This indicates that Vodacom's stock is expected to experience lower and more stable volatility compared to MTN and Telkom in the near future.

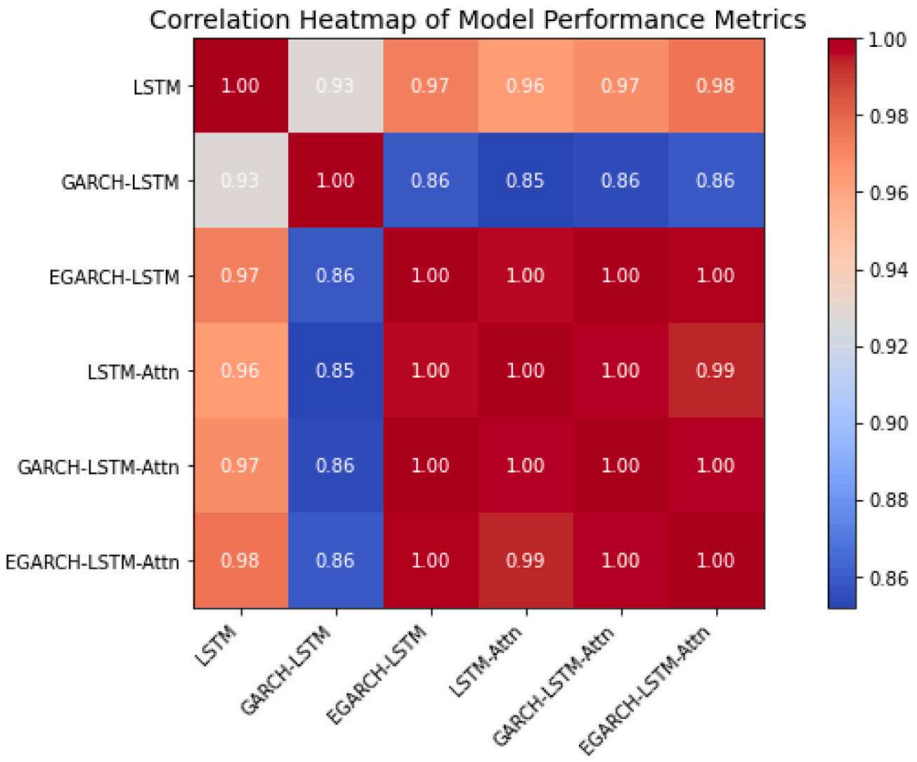

Fig. 7. Correlation Heatmap of DL, Hybrid, and Econometric Models. This heatmap illustrates the correlation between different model performance metrics. DL models (LSTM, LSTM-Attn), hybrid models (GARCH-LSTM, EGARCH-LSTM, and their attention-based variants). Higher correlations (red) indicate strong similarity in performance, while lower correlations (blue) suggest differences in model behavior, with GARCH-LSTM showing the lowest correlation with others. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

GARCH-LSTM and EGARCH-LSTM with and without attention

We compare the performance of six different models across three evaluation metrics which are MAE, MSE, and RMSE. The models were evaluated on three telecommunication stock datasets: MTN, Telkom, and Vodacom. The models considered are LSTM without attention mechanism, LSTM with attention mechanism, GARCH-LSTM without attention mechanism, GARCH-LSTM with attention mechanism, EGARCH-LSTM without attention mechanism and EGARCH-LSTM with attention mechanism. The stock price predictions for MTN (Fig. A.1), Telkom (Fig. A.2), and Vodacom (Fig. A.3) reveal a consistent pattern across different models, highlighting the strengths and weaknesses of each forecasting approach. The LSTM model without attention (blue dashed line) struggles to capture volatility, often underestimating or lagging behind sharp price movements. The GARCH-LSTM and EGARCH-LSTM models (green and red dashed lines) provide moderate improvements by incorporating volatility dynamics, but they still exhibit discrepancies in high-variability regions. In contrast, attention enhanced models (solid lines) LSTM-Attn (blue), GARCH-LSTM-Attn (green), and EGARCH-LSTM-Attn (red) demonstrate superior accuracy across all three stocks. Among them, EGARCH-LSTM-Attn consistently delivers the best predictions, closely aligning with actual prices and effectively capturing rapid fluctuations. This suggests that combining volatility modeling (EGARCH) with attention enhanced LSTMs significantly enhances forecasting accuracy, making it the most effective approach for predicting stock prices in volatile market conditions. To evaluate the performance of the proposed attention-guided hybrid models, we conducted a comparative analysis using three baseline approaches: a standalone LSTM model, a GARCH-LSTM hybrid, and an EGARCH-LSTM hybrid, all implemented both with and without the attention mechanism. These benchmarks enabled us to assess the added value of integrating statistical volatility modeling with attention-enhanced learning.

Fig. 7 presents the correlation heatmap of model performance metrics, illustrating the relationships between different models. High correlation values (close to 1) indicate that the models exhibit similar performance patterns, whereas lower correlation values suggest differences in behavior. The strongest correlations are observed among EGARCH-LSTM, LSTM-Attn, GARCH-LSTM-Attn, and EGARCH-LSTM-Attn, all showing near-perfect similarity (correlation ≈ 1.00), suggesting that these models follow a common performance trend. Conversely, the GARCH-LSTM model exhibits the lowest correlation with others, particularly around 0.85–0.86 when compared to LSTM-Attn, EGARCH-LSTM, and EGARCH-LSTM-Attn. This lower correlation implies that GARCH-LSTM captures distinct performance dynamics, likely because standard GARCH models primarily focus on symmetric volatility and may not effectively capture complex dependencies in financial time series. In contrast, EGARCH-based models account for asymmetric volatility effects, which often provide better adaptability to real-world market fluctuations, leading to their higher correlation with each other. Additionally, attention-based models enhance feature selection by prioritizing critical information, further reinforcing

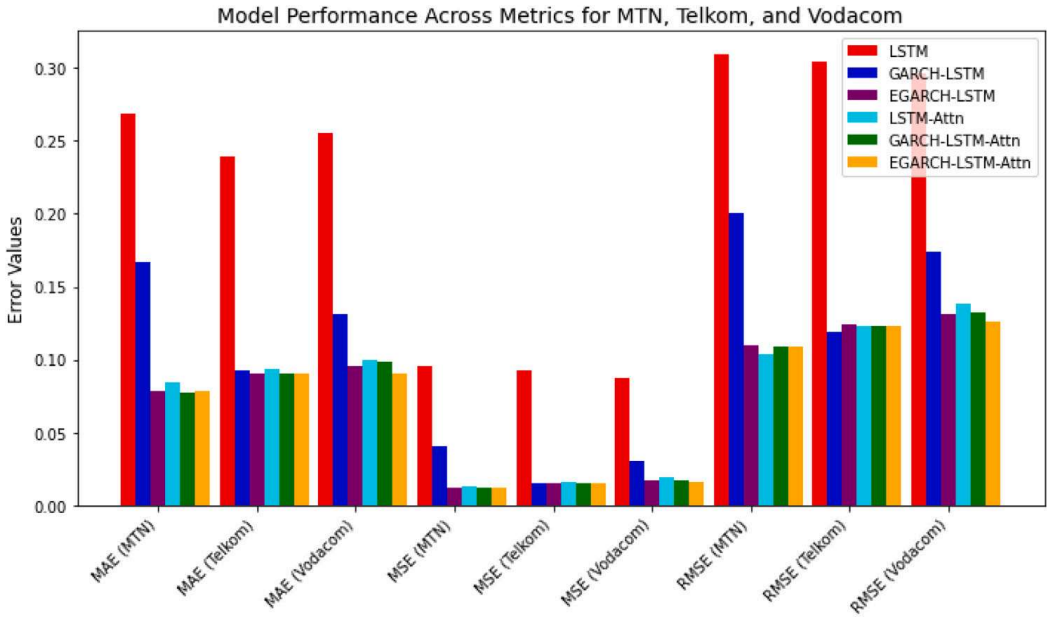

Fig. 8. The bar chart illustrates the performance of different models LSTM, GARCH-LSTM, EGARCH-LSTM, LSTM-Attn, GARCH-LSTM-Attn, and EGARCH-LSTM-Attn across various error metrics for three South African Telecommunication stocks: MTN, Telkom, and Vodacom.

Table 6
Comparison of model evaluation results for LSTM, GARCH-LSTM, and EGARCH-LSTM with and without attention mechanism.

Metric	Without attention			With attention		
	LSTM	GARCH-LSTM	EGARCH-LSTM	LSTM-Attn	GARCH-LSTM-Attn	EGARCH-LSTM-Attn
MAE (MTN)	0.2689	0.1669	0.0783	0.0842	**0.0775**	0.0780
MAE (Telkom)	0.2388	0.0928	0.0904	0.0935	0.0907	**0.0901**
MAE (Vodacom)	0.2556	0.1308	0.0951	0.0992	0.0986	**0.0905**
MSE (MTN)	0.0959	0.0404	0.0120	0.0131	0.0119	**0.0117**
MSE (Telkom)	0.0923	0.0151	0.0153	0.0157	0.0152	**0.0150**
MSE (Vodacom)	0.0879	0.0304	0.0171	0.0193	0.0176	**0.0158**
RMSE (MTN)	0.3097	0.2009	0.1096	**0.1036**	0.1083	0.1093
RMSE (Telkom)	0.3038	**0.1190**	0.1237	0.1231	0.1226	0.1232
RMSE (Vodacom)	0.2965	0.1742	0.1307	0.1388	0.1327	**0.1259**

their strong correlation. Overall, the heatmap suggests that models incorporating EGARCH and attention mechanisms yield highly consistent performance, whereas GARCH-LSTM stands out as the most distinct model due to its limitations in capturing asymmetric volatility and long-range dependencies effectively.

Fig. 8 presents a comparison of the evaluation measures for the models applied to MTN, Telkom, and Vodacom stocks. The LSTM model (red bars) exhibits the highest MSE across all stocks, indicating relatively poor performance compared to the other models, as shown in Fig. 8. Both the GARCH-LSTM and EGARCH-LSTM models (blue and green bars) show lower MSE values, suggesting that the integration of GARCH components helps improve the capture of volatility. The attention-based models (cyan, dark blue, and orange bars) further enhance the performance, achieving the lowest MSE values. This trend is consistent across the different models, where the LSTM model performs the worst with respect to MSE. The GARCH-LSTM and EGARCH-LSTM models outperform the LSTM, and the attention-based models consistently deliver the best results, highlighting the benefit of incorporating attention mechanisms in time series forecasting. This pattern of performance remains consistent across all three stocks, with attention-based models continuing to outperform other configurations, as shown in Table 6. Among the three stocks, Vodacom exhibits the highest error values, suggesting that predicting its volatility is more challenging compared to MTN and Telkom. The evaluation of the LSTM, GARCH-LSTM, and EGARCH-LSTM models on MTN, Telkom, and Vodacom stock data reveals significant differences in predictive accuracy. The LSTM model consistently shows the highest error values, reflecting its inability to effectively capture the volatility in stock prices. In contrast, incorporating GARCH and EGARCH components into the LSTM architecture reduces both the MAE and MSE, highlighting the advantages of integrating volatility modeling with DL. Additionally, the inclusion of an attention mechanism further enhances the model's predictive performance, as evidenced by the lower error values across all models. Specifically, the EGARCH-LSTM model with attention achieves the best results, outperforming all other configurations. This suggests that the EGARCH model's capacity to capture asymmetric volatility effects, when combined with the ability of the attention mechanism to concentrate on relevant time steps, significantly improves forecasting accuracy for stock prices. Among the stocks analyzed, Vodacom shows the highest error values, indicating that it is the most difficult to predict. This could be due to its

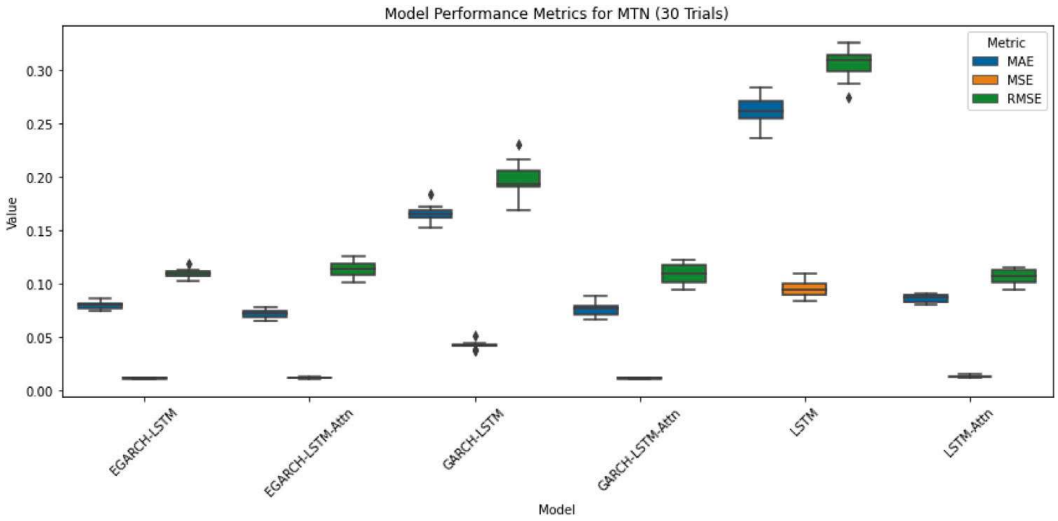

Fig. 9. Boxplot comparison of model performance across MTN stock. The models evaluated include LSTM, LSTM with Attention (LSTM-Attn), GARCH-LSTM, GARCH-LSTM-Attn, EGARCH-LSTM, and EGARCH-LSTM-Attn. The metrics shown are MAE, MSE, and RMSE over 30 trials. EGARCH-based hybrids achieve consistently lower error values, indicating their robustness in modeling stock volatility.

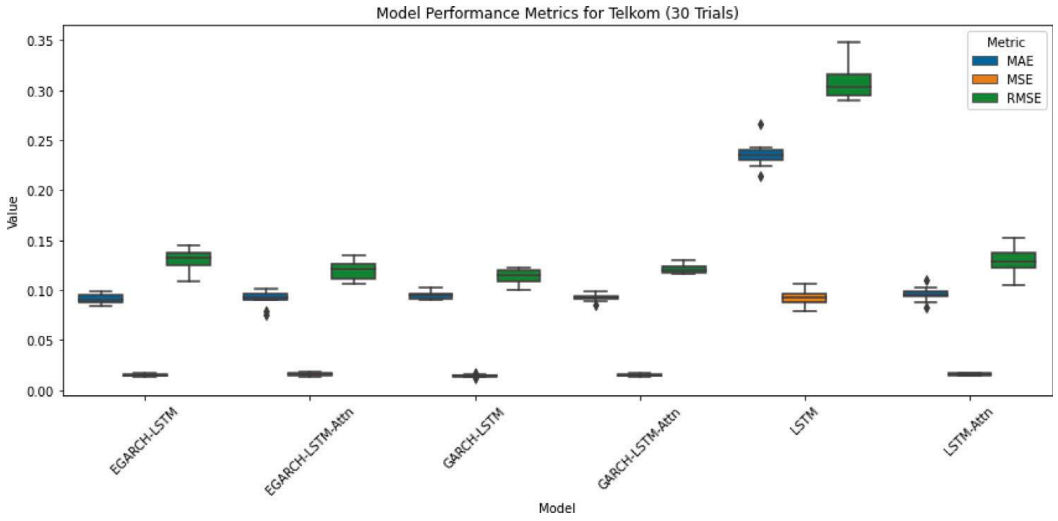

Fig. 10. Boxplot comparison of model performance across Telkom stock. The models evaluated include LSTM, LSTM with Attention (LSTM-Attn), GARCH-LSTM, GARCH-LSTM-Attn, EGARCH-LSTM, and EGARCH-LSTM-Attn. The metrics shown are MAE, MSE, and RMSE over 30 trials. EGARCH-based hybrids achieve consistently lower error values, indicating their robustness in modeling stock volatility.

higher volatility or more complex market dynamics. Overall, these findings support the conclusion that hybrid models that integrate volatility components and attention mechanisms offer superior forecasting accuracy compared to traditional LSTM models.

To provide additional statistical robustness, Tables 7, 8, and 9 report 95% confidence intervals alongside the mean and standard deviation of each model's performance metric (MAE, MSE, RMSE) on the MTN, Telkom, and Vodacom stocks. These intervals offer insight into the reliability and variability of the model estimates across repeated trials. Notably, the EGARCH-LSTM-Attn model demonstrates the narrowest confidence intervals, indicating greater stability and consistency in its predictions. In addition to numerical metrics, Fig. 9, 10 and 11 visualizes the distribution of errors using box plots across all models and metrics. This visualization highlights the spread and outliers in each model's performance. The LSTM model shows wider interquartile ranges and more extreme outliers, while the EGARCH-LSTM-Attn model demonstrates tighter distributions, consistent with its lower variance and confidence interval width.

Discussion

The results of this work reinforce the importance of integrating volatility models and attention mechanisms in DL-based financial forecasting. Traditional LSTMs struggle to model stock price fluctuations effectively, as indicated by their consistently high error values. The incorporation of GARCH and EGARCH components enhances predictive accuracy by accounting for heteroskedasticity

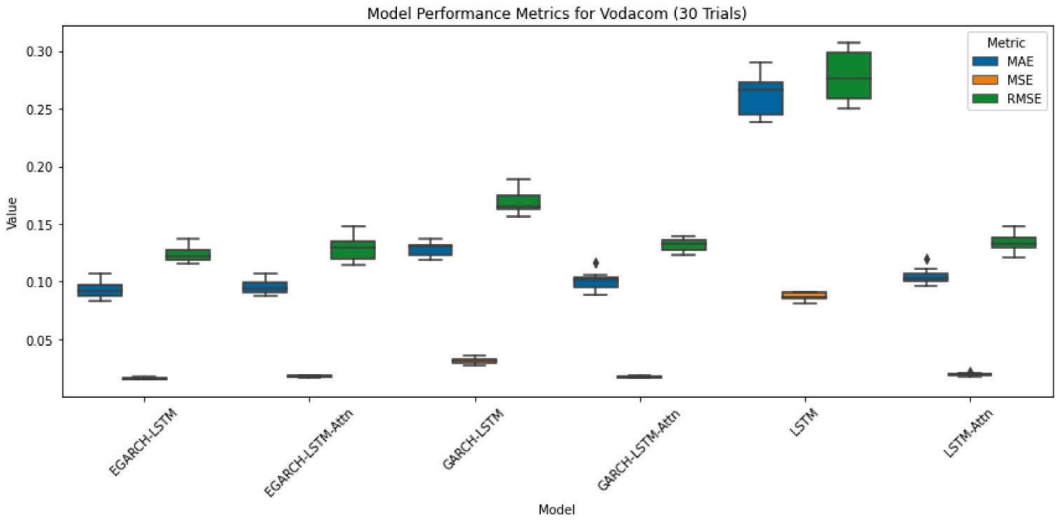

Fig. 11. Boxplot comparison of model performance across Vodacom stock. The models evaluated include LSTM, LSTM with Attention (LSTM-Attn), GARCH-LSTM, GARCH-LSTM-Attn, EGARCH-LSTM, and EGARCH-LSTM-Attn. The metrics shown are MAE, MSE, and RMSE over 30 trials. EGARCH-based hybrids achieve consistently lower error values, indicating their robustness in modeling stock volatility.

Table 7
Comparison of model performance on MTN stock volatility prediction using MAE, MSE, and RMSE. Lower values indicate better model performance. Results include mean error values with standard deviation and 95% confidence intervals.

Metric	Stock	Model	Mean	StdDev	CI lower	CI upper
MAE	MTN	EGARCH-LSTM-Attn	0.076831	0.005852	0.074737	0.078925
MAE	MTN	EGARCH-LSTM	0.077810	0.005563	0.075819	0.079801
MAE	MTN	GARCH-LSTM-Attn	0.078281	0.005809	0.076202	0.080360
MAE	MTN	LSTM-Attn	0.085093	0.005861	0.082996	0.087190
MAE	MTN	GARCH-LSTM	0.168152	0.010738	0.164309	0.171995
MAE	MTN	LSTM	0.265359	0.016941	0.259297	0.271421
MSE	MTN	GARCH-LSTM-Attn	0.011560	0.000715	0.011304	0.011816
MSE	MTN	EGARCH-LSTM	0.011886	0.000968	0.011540	0.012232
MSE	MTN	EGARCH-LSTM-Attn	0.012107	0.000819	0.011814	0.012400
MSE	MTN	LSTM-Attn	0.013303	0.000776	0.013025	0.013581
MSE	MTN	GARCH-LSTM	0.041045	0.002808	0.040040	0.042050
MSE	MTN	LSTM	0.095764	0.006101	0.093581	0.097947
RMSE	MTN	LSTM-Attn	0.105309	0.006796	0.102877	0.107741
RMSE	MTN	GARCH-LSTM-Attn	0.107714	0.007652	0.104976	0.110452
RMSE	MTN	EGARCH-LSTM	0.108029	0.007060	0.105503	0.110555
RMSE	MTN	EGARCH-LSTM-Attn	0.108280	0.007912	0.105449	0.111111
RMSE	MTN	GARCH-LSTM	0.201555	0.015516	0.196003	0.207107
RMSE	MTN	LSTM	0.311122	0.022823	0.302955	0.319289

in financial time series data. Notably, EGARCH-LSTM outperforms GARCH-LSTM, suggesting that EGARCH's ability to model asymmetric volatility provides a more realistic representation of market conditions. The impact of the attention mechanism is particularly noteworthy. Across all model configurations, attention-based variants consistently yield lower error metrics, confirming the effectiveness of attention in improving model interpretability and accuracy. This is because attention enables the model to focus on the most relevant historical data points, rather than treating all past information equally. The combination of EGARCH and attention achieves the best results, highlighting the synergy between capturing volatility patterns and selectively emphasizing critical time steps. Stock-specific analysis reveals that Vodacom is the most challenging stock to predict, with consistently higher error values compared to MTN and Telkom. This could be attributed to greater market fluctuations, external economic factors, or lower trading liquidity. Conversely, MTN demonstrates relatively lower forecasting errors, suggesting more predictable price movements. These findings highlight the importance of adapting forecasting models to the characteristics of individual stocks. It is important to acknowledge that the one-year historical dataset used in this study may not fully capture the impact of rare or extreme events, such as pandemics, natural disasters, or geopolitical shocks. These events can introduce substantial, sudden volatility that standard market patterns do not reflect. As such, the forecasting models may exhibit different performance dynamics under such conditions. We recognize this limitation and suggest that future research extend the temporal range of the dataset to include multiple years and diverse market regimes. Doing so would allow for a more comprehensive evaluation of model robustness under extreme market conditions. The broader implications of this study suggest that hybrid DL models incorporating domain-specific knowledge, such as volatility modeling and attention mechanisms, are more effective than traditional approaches. As financial

Table 8
Model performance metrics for Telkom stock volatility forecasting. Evaluation metrics include MAE, MSE, and RMSE with their means, standard deviations, and 95% confidence intervals.

Metric	Stock	Model	Mean	StdDev	CI lower	CI upper
MAE	Telkom	EGARCH-LSTM	0.090654	0.005741	0.088600	0.092708
MAE	Telkom	EGARCH-LSTM-Attn	0.091674	0.006295	0.089421	0.093927
MAE	Telkom	GARCH-LSTM-Attn	0.093321	0.005273	0.091434	0.095208
MAE	Telkom	GARCH-LSTM	0.094631	0.005049	0.092824	0.096438
MAE	Telkom	LSTM-Attn	0.094892	0.006340	0.092623	0.097161
MAE	Telkom	LSTM	0.236775	0.015564	0.231205	0.242345
MSE	Telkom	EGARCH-LSTM	0.015062	0.001121	0.014661	0.015463
MSE	Telkom	GARCH-LSTM-Attn	0.015086	0.000976	0.014737	0.015435
MSE	Telkom	GARCH-LSTM	0.015130	0.001061	0.014750	0.015510
MSE	Telkom	EGARCH-LSTM-Attn	0.015613	0.001192	0.015186	0.016040
MSE	Telkom	LSTM-Attn	0.015665	0.001105	0.015270	0.016060
MSE	Telkom	LSTM	0.091686	0.006617	0.089318	0.094054
RMSE	Telkom	GARCH-LSTM	0.117352	0.008294	0.114384	0.120320
RMSE	Telkom	EGARCH-LSTM-Attn	0.122007	0.007730	0.119241	0.124773
RMSE	Telkom	GARCH-LSTM-Attn	0.123973	0.006803	0.121539	0.126407
RMSE	Telkom	EGARCH-LSTM	0.124855	0.009407	0.121489	0.128221
RMSE	Telkom	LSTM-Attn	0.125306	0.008879	0.122129	0.128483
RMSE	Telkom	LSTM	0.304822	0.022065	0.296926	0.312718

Table 9
Evaluation of hybrid and DL models for Vodacom stock volatility prediction. Metrics reported are MAE, MSE, and RMSE with corresponding means, standard deviations, and 95% confidence intervals.

Metric	Stock	Model	Mean	StdDev	CI lower	CI upper
MAE	Vodacom	EGARCH-LSTM	0.092600	0.006629	0.090228	0.094972
MAE	Vodacom	EGARCH-LSTM-Attn	0.097120	0.006343	0.094850	0.099390
MAE	Vodacom	GARCH-LSTM-Attn	0.099173	0.009325	0.095836	0.102510
MAE	Vodacom	LSTM-Attn	0.102098	0.006407	0.099805	0.104391
MAE	Vodacom	GARCH-LSTM	0.128909	0.006157	0.126706	0.131112
MAE	Vodacom	LSTM	0.255831	0.017749	0.249480	0.262182
MSE	Vodacom	EGARCH-LSTM	0.015758	0.001056	0.015380	0.016136
MSE	Vodacom	EGARCH-LSTM-Attn	0.017260	0.001196	0.016832	0.017688
MSE	Vodacom	GARCH-LSTM-Attn	0.017580	0.001376	0.017088	0.018072
MSE	Vodacom	LSTM-Attn	0.019242	0.001401	0.018741	0.019743
MSE	Vodacom	GARCH-LSTM	0.030278	0.002530	0.029373	0.031183
MSE	Vodacom	LSTM	0.089615	0.005829	0.087529	0.091701
RMSE	Vodacom	EGARCH-LSTM	0.126122	0.006638	0.123747	0.128497
RMSE	Vodacom	EGARCH-LSTM-Attn	0.130183	0.011395	0.126105	0.134261
RMSE	Vodacom	GARCH-LSTM-Attn	0.132126	0.009124	0.128861	0.135391
RMSE	Vodacom	LSTM-Attn	0.137605	0.009795	0.134100	0.141110
RMSE	Vodacom	GARCH-LSTM	0.170093	0.009808	0.166583	0.173603
RMSE	Vodacom	LSTM	0.293523	0.024045	0.284919	0.302127

markets become increasingly complex, integrating statistical models with DL frameworks can enhance accuracy and improve the robustness of predictions, benefiting both researchers and practitioners in finance.

Conclusion

This work proposed a novel attention-guided hybrid volatility forecasting framework that combines econometric models (GARCH, EGARCH) with DL models (LSTM) to improve predictive accuracy for South African telecommunication stocks (MTN, Telkom, and Vodacom). By incorporating an attention mechanism, the model dynamically focuses on critical time steps, enhancing both accuracy and interpretability of volatility forecasts. Our results demonstrate that hybrid models consistently outperform standalone LSTM models, with attention-based variants further reducing forecasting errors, particularly in terms of MAE and MSE. EGARCH-LSTM-Attn achieved the lowest MSE across all stocks (0.0117 for MTN, 0.0150 for Telkom, and 0.0158 for Vodacom) and strong MAE performance, while RMSE improvements were model-specific: LSTM-Attn performed best for MTN (0.1036), GARCH-LSTM for Telkom (0.1190), and EGARCH-LSTM-Attn for Vodacom (0.1259). These findings underscore the effectiveness of attention-guided hybrid models for volatility forecasting in emerging markets, where external shocks often influence market dynamics. Future research should explore longer historical datasets to capture extreme events, integrate macroeconomic and sentiment indicators, investigate advanced architectures such as multi-head attention and Transformers, and evaluate these models in live or simulated trading environments to assess real-world applicability and robustness.

CRediT authorship contribution statement

Wandile Nhlapho: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Visualisation, Writing – original draft, Writing – review & editing. **Marcellin Atemkeng:** Conceptualization, Methodology, Visualisation, Formal analysis, Resources, Writing – review & editing, Validation. **Jean-Claude Ndogmo:** Conceptualization, Formal analysis, Visualisation, Data curation, Resources, Writing – review & editing, Supervision, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

This section contains additional materials provide further insights into the methodology, results, and comparisons of the models used in this work, enriching the understanding of financial volatility forecasting and the practical application of hybrid modeling techniques.

Table A.1
Advantages and limitations of GARCH, EGARCH, LSTM, GARCH-LSTM, and EGARCH-LSTM models.

Model	Pros	Cons
GARCH	Captures volatility clustering; interpretable parameters	Assumes symmetric shocks; sensitive to non-stationarity
EGARCH	Captures asymmetric volatility; more flexible than GARCH	Computationally complex; requires more parameters
LSTM	Learns non-linear patterns; captures long-term dependencies	Computationally expensive; prone to overfitting
GARCH-LSTM	Combines GARCH and LSTM strengths for accurate forecasts	High computational cost; complex hyperparameter tuning
EGARCH-LSTM	Captures asymmetric volatility and non-linear dependencies	Challenging to optimize; computationally intensive

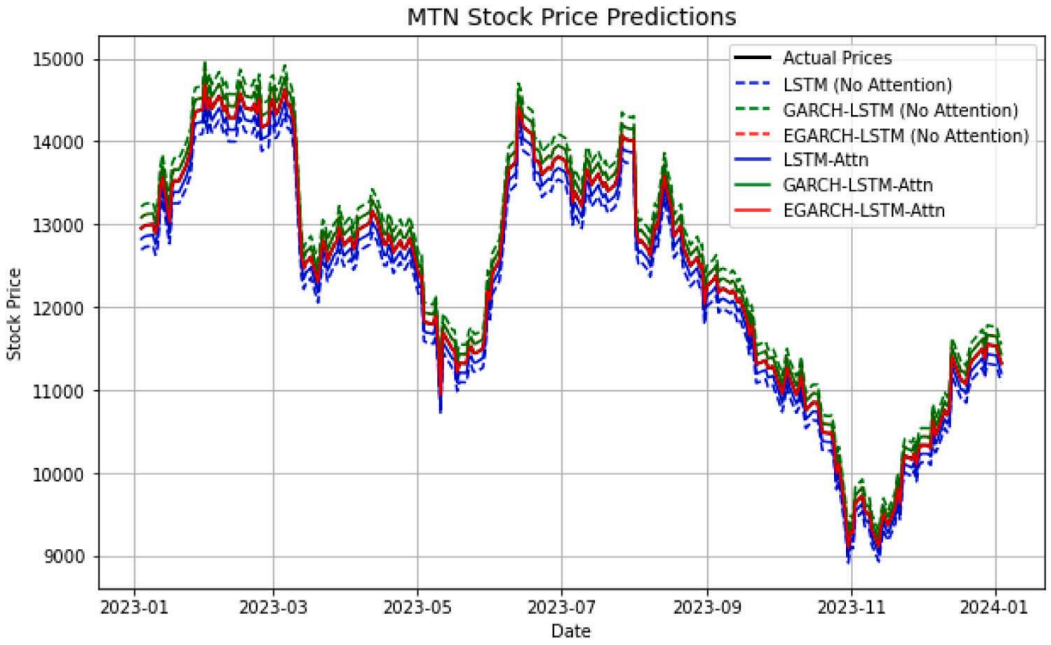

Fig. A.1. MTN stock: Actual vs. Predicted prices using LSTM, GARCH-LSTM, and EGARCH-LSTM with and without attention mechanism models.

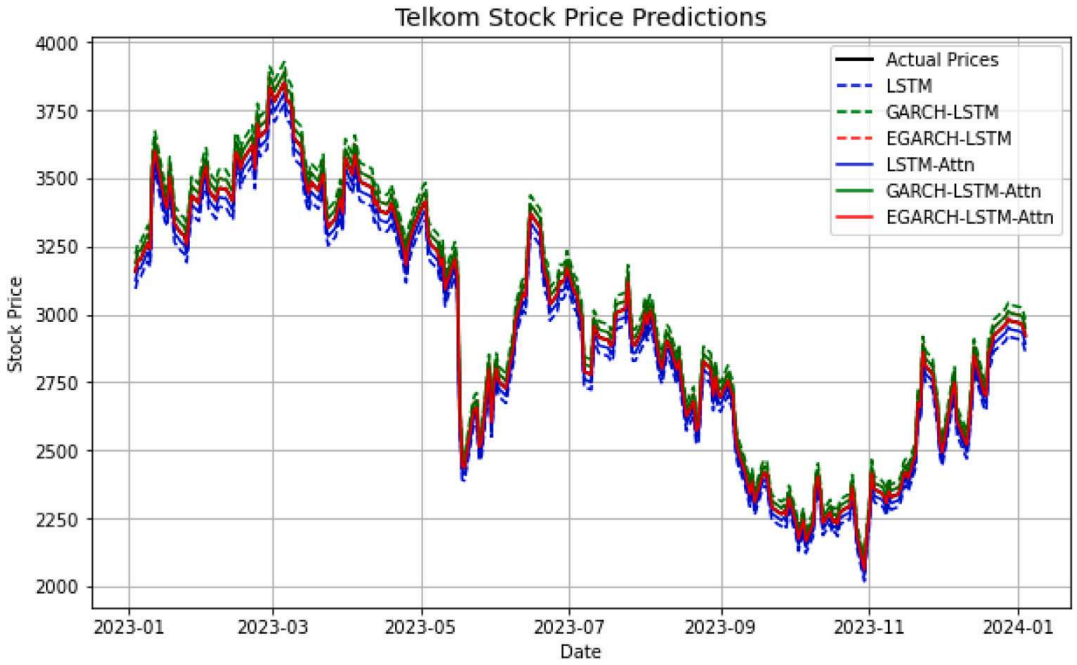

Fig. A.2. Telkom stock: Actual vs. Predicted prices using LSTM, GARCH-LSTM, and EGARCH-LSTM with and without attention mechanism models.

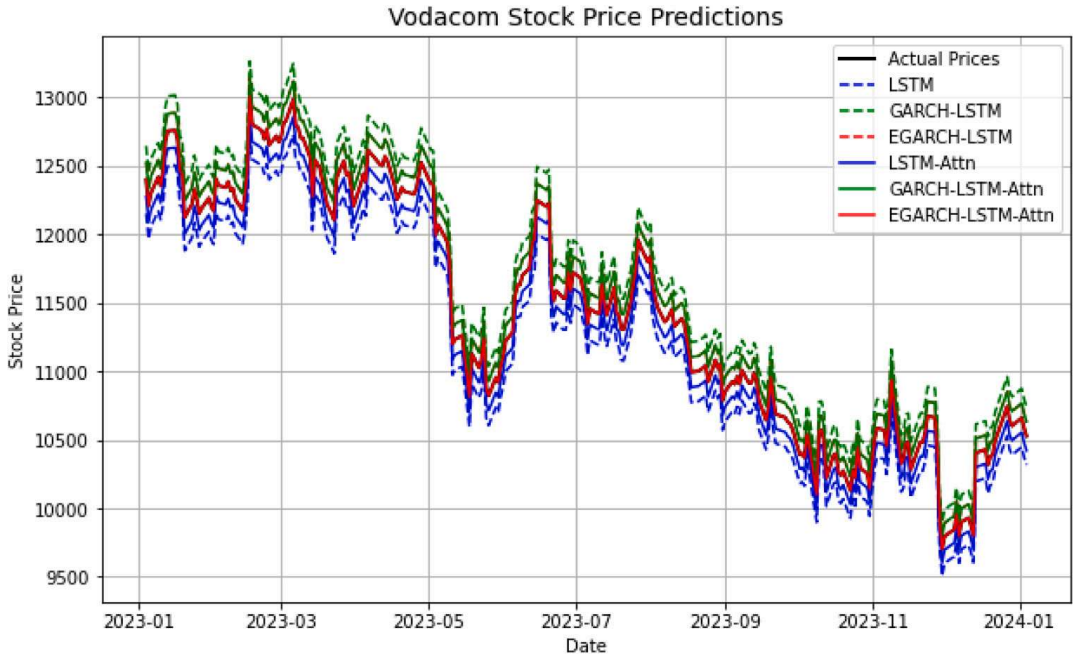

Fig. A.3. Vodacom stock: Actual vs. Predicted prices using LSTM, GARCH-LSTM, and EGARCH-LSTM with and without attention mechanism models.

Table A.2
Studies on hybrid models in financial volatility forecasting.

Study title	Authors	Model used	Key findings
The hybrid forecast of S&P 500 volatility ensembled from VIX, GARCH, and LSTM models [27]	Roszyk et al (2024)	GARCH-LSTM	The hybrid model outperforms standalone models in forecasting S&P 500 volatility by incorporating VIX, GARCH, and LSTM models.
From GARCH to Neural Network for Volatility Forecast [28]	Pengfei Zhao et al (2024)	GARCH-NN	Establishes an equivalence between GARCH models and neural networks, proposing a GARCH-NN approach that enhances volatility forecasting by integrating GARCH parameters into neural networks.
Forecasting high-dimensional covariance matrices of asset returns with hybrid GARCH-LSTMs [23]	Lucien Boulet (2021)	GARCH-LSTM	The hybrid model enhances risk modeling in portfolio management by effectively forecasting high-dimensional covariance matrices.
Combining deep learning and GARCH models for financial volatility and risk forecasting [25]	Jakub Michańków et al (2023)	GARCH-GRU	Combining GARCH models with GRU neural networks improves volatility and risk forecasting for assets like S&P 500, gold, and Bitcoin.
Forecasting commodity market returns volatility: A hybrid ensemble learning GARCH-LSTM based approach [24]	Kshitij Kakade et al (2022)	GARCH-LSTM, EGARCH-LSTM	Integrating forecasts from multiple GARCH-type models into a hybrid LSTM model leads to superior volatility forecasting for commodities like nickel, copper, tin, lead, and gold.
Forecasting the volatility of stock price index: A hybrid model integrating LSTM with multiple GARCH-type models [22]	Kim et al (2018)	GARCH-LSTM	The proposed GEW-LSTM model (a hybrid of LSTM and three GARCH-type models) outperforms standalone models like GARCH, EGARCH, and deep feedforward neural networks (DFN).
GARCH-Informed neural networks for volatility prediction in financial markets [29]	Xu et al. (2024)	GARCH-LSTM (GINN)	Introduced GARCH-Informed Neural Networks (GINN), combining GARCH with LSTM to improve volatility prediction accuracy over standalone models.
Forecasting stock volatility via Hybrid DL and GARCH family models: A Case Study from BIST30 [30]	Ulu (2025)	GARCH-LSTM, GARCH-BiLSTM	Demonstrated that hybrid models outperform standalone models in forecasting stock volatility in the Turkish BIST30 index, especially during the COVID-19 period.
Bridging econometrics and AI: VaR estimation via reinforcement learning and GARCH models [31]	Pokou et al. (2025)	GARCH-DDQN	Proposed a hybrid framework combining GARCH models with Deep Q-Networks for improved Value-at-Risk estimation, showing enhanced accuracy and reduced capital requirements.
Hybrid deep learning and GARCH-Family models for forecasting volatility of cryptocurrencies [26]	Amirshahi et al (2023)	GARCH-DFFNN, GARCH-LSTM	Found that integrating GARCH-type models with DL architectures enhances the predictive power for cryptocurrency volatility forecasting.

Table A.3
Forecasted volatility for MTN, Telkom, and Vodacom for the next 10 days (GARCH).

Day	MTN forecasted volatility	Telkom forecasted volatility	Vodacom forecasted volatility
h.01	1.8419	2.6594	1.6029
h.02	1.9030	2.7655	1.6045
h.03	1.9077	2.8384	1.6062
h.04	1.9081	2.8889	1.6078
h.05	1.9081	2.9242	1.6095
h.06	1.9081	2.9489	1.6111
h.07	1.9081	2.9663	1.6128
h.08	1.9081	2.9786	1.6144
h.09	1.9081	2.9873	1.6161
h.10	1.9081	2.9934	1.6177

Table A.4
Forecasted volatility for MTN, Telkom, and Vodacom for the next 10 days (EGARCH).

Day	MTN forecasted volatility	Telkom forecasted volatility	Vodacom forecasted volatility
h.01	0.0184	0.0249	0.0115
h.02	0.0191	0.0267	0.0115
h.03	0.0192	0.0282	0.0115
h.04	0.0192	0.0290	0.0115
h.05	0.0192	0.0297	0.0114
h.06	0.0193	0.0301	0.0114
h.07	0.0192	0.0304	0.0115
h.08	0.0192	0.0305	0.0114
h.09	0.0192	0.0305	0.0114
h.10	0.0192	0.0306	0.0114

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